AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT	1. CONTRACT ID CODE	PAGE OF PAGES 1 | 2

2. AMENDMENT/MODIFICATION NO.	3. EFFECTIVE	4. REQUISITION/PURCHASE	5. PROJECT NO. (If applicable)
W9127825RA015-0002	02 JUNE 2025		CHC24010

6. ISSUED BY CODE	7. ADMINISTERED BY(If other than item 6) CODE
Corps of Engineers 109 St. Joseph St. Mobile, AL 36602	

8. NAME AND ADDRESS OF CONTRACTOR (No., street, county, State and ZIP code)

CODE | FACILITY CODE

☒ 9A. AMENDMENT OF SOLICITATION NO.
W9127825RA015

9B. DATED (SEE ITEM 11)

24 APRIL 2025

☐ 10A. MODIFICATION OF CONTRACT/ORDER NO.

10B. DATED (SEE ITEM 13)

11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS

☒ The above numbered solicitation is amended as set forth in item 14. The hour and date specified for receipt of Offers ☒ is extended, ☐ is not extended. Offers must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended, by one of the following methods: (a) By completing items 8 and 15, and returning copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or telegram which includes a reference to the solicitation and amendment numbers. **FAILURE OF YOUR ACKNOWLEDGEMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER.** If by virtue of this amendment you desire to change an offer already submitted, such change may be made by telegram or letter, provided each telegram or letter makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.

12. ACCOUNTING AND APPROPRIATION DATA *(if required)*

13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS, IT MODIFIES THE CONTRACT/ORDER NO. AS DESCRIBED IN ITEM 14.

☐ A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NO. IN ITEM 10A

☐ B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation date, etc.) SET FORTH IN ITEM 14, PURSUANT TO AUTHORITY OF FAR 43.103(b)

☐ C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:

☐ D. OTHER (Specify type of modification and authority)

E. IMPORTANT: Contractor ☐ is not, ☐ is required to sign this document and return copies to the issuing office.

14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible)

The subject solicitation for: HOLT LOCK MONOLITH R14 RESTORATION - HOLT DAM BLACK WARRIOR RIVER, ALABAMA (TUSCALOOSA COUNTY)

Is modified in the following: REFER TO THE ENCLOSED REVISED SPECIFICATION/DRAWING REVISIONS FOR AMENDMENT NO. 0002
NOTE: THE RECEIPT OF PROPOSAL DATE IS HEREBY REVISED BY THIS AMENDMENT. REFER TO THE ENCLOSED SF1442, SOLICITATION, OFFER & AWARD FORM.

Except as provided herein, all terms and conditions of the document reference in item 9A or 10A, as Heretofore changed, remains unchanged and in full force and effect.

15A. NAME AND TITLE OF SIGNER (Type or print)		16A. NAME AND TITLE OF CONTRACTING OFFICE (Type or print)	
15B. CONTRACTOR/OFFEROR (Signature of person authorized to sign)	15C. DATE SIGNED	16B. UNITED STATES OF AMERICA BY (Signature of contracting officer)	16C. DATE SIGNED

PART I-REVISIONS MADE BY ADDED AND/OR REPLACEMENT PARAGRAPHS/PAGES/SECTIONS

The items listed below are to be replaced by the corresponding added and/or revised paragraphs/pages or sections. Added and/or revised paragraphs/pages or sections are indicated by a note in bottom right hand corner of each paragraph or page. Added sections are hereby made a part of the contract and are to be inserted in the specification in the proper numerical/alphabetical sequence.

Within the specifications, deletions from the specifications are indicated by strikethrough, e.g.: ~~deletions are marked with strikethrough~~ and additions to the specifications including revisions/substitutions are indicated in bold, italic and underlined, e.g.: ***additions are indicated thus.***

SECTION	Corresponding Added or Revised Paragraph Page, and/or Section
SF1442	Revised box 13.a as indicated herein.
Bidding Schedule	Replaced in its entirety.
Explanation of Bid Items	Replaced in its entirety.
Technical Specifications	
01 00 00	Revised paragraph 1.1 as indicated herein.
13 51 00.10	Revised paragraph 3.1 as indicated herein.
31 68 13	Revised paragraph 1.7.7, 3.1, 3.2.1.1, 3.2.1.5 and 3.2.5.1 as indicated herein.

PART II - NOTE: Revised, replacement and added drawings are listed below. These revised, replacement and added drawings are to be inserted into the folio in the proper numerical sequence. Drawings that have been revised or replaced by this amendment shall be deleted from the folio. All drawings listed below are revised unless indicated otherwise.

Sheet Reference Number	Title
G-001	COVER SHEET
G-002	GENERAL NOTES, ABBREVIATIONS
S-113	14R CULVERT CRACK LOCATIONS
S-131	ANCHOR LOCATION ELEVATION AND LENGTH/STRESSING SEQUENCE TABLE
S-402	TOE BLOCK DETAILS

Encl as stated
Revised pages of the specifications as indicated in Part I.
05 Revised drawings as indicated in Part II.

SOLICITATION, OFFER AND AWARD *(Construction, Alteration, or Repair)*	1. SOLICITATION NUMBER W9127825RA015	2. TYPE OF SOLICITATION ☐ SEALED BID (IFB) INVITATION FOR BID ☒ NEGOTIATED (RFP) REQUEST FOR PROPOSAL	3. DATE ISSUED 24 APR 2025	PAGE 1 OF PAGES 2

IMPORTANT - The "offer" section on the reverse must be fully completed by offeror.

4. CONTRACT NO.	5. REQUISITION/PURCHASE REQUEST NO.	6. PROJECT NUMBER CHC24010

7. ISSUED BY CODE CT	8. ADDRESS OFFER TO
U.S. ARMY ENGINEER DISTRICT, MOBILE CONTRACTING DIVISION (CESAM-CT)	SAME AS BLOCK 7

9. FOR INFORMATION CALL	a. NAME Jennifer Suit / Elizabeth (Lisa) Trimble	b. TELEPHONE NO. (*Include area code*) (*NO COLLECT CALLS*) (251) 441-6466 / (251) 690-3329

SOLICITATION

NOTE: In sealed bid solicitations "offer" and "offeror" mean "bid" and "bidder".

10. THE GOVERNMENT REQUIRES PERFORMANCE OF THE WORK DESCRIBED IN THESE DOCUMENTS (*Title, identifying no., date*):

HOLT LOCK MONOLITH R14 RESTORATION – HOLT DAM BLACK WARRIOR RIVER, ALABAMA (TUSCALOOSA COUNTY)

* See Section 01 00 00, Paragraph "COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK".
**Block 13A. : Refer to Section 00 11 00 for the number of copies to submit with the original offer.

11. The contractor shall begin performance within 8 calendar days and complete it within * calendar days after receiving ☐ award, ☒ notice to proceed. This performance period is ☒ mandatory, ☐ negotiable. (See ______ .)

12a. THE CONTRACTOR MUST FURNISH ANY REQUIRED PERFORMANCE AND PAYMENT BONDS? (*If "YES," indicate within how many calendar days after award in Item 12b.)* ☒ YES ☐ NO	12b. CALENDAR DAYS 10

13. ADDITIONAL SOLICITATION REQUIREMENTS:

a. Sealed offers in original and ** copies to perform the work required are due at the place specified in Item 8 by 1400 (***hour***) local time ~~28 MAY 04~~ ***10 JUNE*** 2025 (date). If this is a sealed bid solicitation, offers must be publicly opened at that time. Sealed envelopes containing offers shall be marked to show the offeror's name and address, the solicitation number, and the date and time offers are due.

b. An offer guarantee ☒ is, ☐ is not required.

c. All offers are subject to the (1) work requirements, and (2) other provisions and clauses incorporated in the solicitation in full text or by reference.

d. Offers providing less than 120 calendar days for Government acceptance after the date offers are due will not be considered and will be rejected.

OFFER (*Must be fully completed by offeror*)

14. NAME AND ADDRESS OF OFFEROR (*Include ZIP Code*)	15. TELEPHONE NO. (*Include area code*)
	16. REMITTANCE ADDRESS (*Include only if different than Item 14*.)
CODE FACILITY CODE	

17. The offeror agrees to perform the work required at the prices specified below in strict accordance with the terms of this solicitation, if this offer is accepted by the Government in writing within _________ calendar days after the date offers are due. (*Insert any number equal to or greater than the minimum requirement stated in Item 13d. Failure to insert any number means the offeror accepts the minimum in Item 13d.)*

AMOUNTS ▶

18. The offeror agrees to furnish any required performance and payment bonds.

19. ACKNOWLEDGMENT OF AMENDMENTS

(*The offeror acknowledges receipt of amendments to the solicitation - give number and date of each*)

AMENDMENT NUMBER										
DATE										

20a. NAME AND TITLE OF PERSON AUTHORIZED TO SIGN OFFER (*Type or print*)	20b. SIGNATURE	20c. OFFER DATE

AWARD (*To be completed by Government*)

21. ITEMS ACCEPTED:

22. AMOUNT	23. ACCOUNTING AND APPROPRIATION DATA

24. SUBMIT INVOICES TO ADDRESS SHOWN IN (4 copies unless otherwise specified) ▶	ITEM	25. OTHER THAN FULL AND OPEN COMPETITION PURSUANT TO ☐ 10 U.S.C. 3204(a) ( ) ☐ 41 U.S.C. 3304(a) ( )

26. ADMINISTERED BY CODE		27. PAYMENT WILL BE MADE BY

CONTRACTING OFFICER WILL COMPLETE ITEM 28 OR 29 AS APPLICABLE

☐ 28. NEGOTIATED AGREEMENT (*Contractor is required to sign this document and return _______ copies to issuing office*.) Contractor agrees to furnish and deliver all items or perform all work, requisitions identified on this form and any continuation sheets for the consideration stated in this contract. The rights and obligations of the parties to this contract shall be governed by (a) this contract award, (b) the solicitation, and (c) the clauses, representations, certifications, and specifications incorporated by reference in or attached to this contract.

☐ 29. AWARD (*Contractor is not required to sign this document.*) Your offer on this solicitation is hereby accepted as to the items listed. This award consummates the contract, which consists of (a) the Government solicitation and your offer, and (b) this contract award. No further contractual document is necessary.

30a. NAME AND TITLE OF CONTRACTOR OR PERSON AUTHORIZED TO SIGN (*Type or print*)		31a. NAME OF CONTRACTING OFFICER (*Type or print*)	
30b. SIGNATURE	30c. DATE	31b. UNITED STATES OF AMERICA BY	31c. AWARD DATE

REVISED BY AMENDMENT NO. W9127825RA015-0002

BIDDER'S NAME:____

BID SCHEDULE

Item No.	Description	Estimated Quantity	Unit	Unit Price	Amount
Base Bid:					
1	Site Work	1	JOB	XXX	___________
2	Marine Plant (Tugboat and Barge)	1	JOB	XXX	___________
3	Intermediate Mobilization and Demobilization	___	EA	___	___________
4	Horizontal 7” Core Drill for Thru-Bolt Anchor	2,612	LF	___	___________
5	Inclined 8” Core Drill For Embedded Anchor	3,036	LF	___	___________
6	Inclined 9” Core Drill For Toe Block Embedded Anchor	750	LF	___	___________
7	Monolith Monitoring	1	JOB	XXX	___________
8	Anchor End Recess	184	EA	___	___________
9	2.5” Diameter Anchor Install for Thru-Bolt Anchorage	64	EA	___	___________
10	2.5” Diameter Anchor Install for Embedded Anchorage	67	EA	___	___________
11	3” Diameter Anchor Install for Embedded Anchorage At Toe Block	15	EA	___	___________
12	Cementitious *Grouting*	1200	CFT	___	___________
13	Re-Drilling	2000	LFT	___	___________
14	Water Stop	1	JOB	XXX	___________
15	Lock Dewatering	1	JOB	XXX	___________
16	Epoxy Grout Injection of cracks	310	GAL	___	___________
17	Concrete Toe Block	1	JOB	XXX	___________

REPLACED BY AMENDMENT NO. W9127825RA015-0002

18	Polyurethane Grout Injection of cracks	450	GAL	___	___________
19	MISC. Concrete Work	1	JOB	___	___________

TOTAL BASE BID ___________

Bid Option:

20	Remove Riverside Filling Valve	1	JOB	XXX	___________

TOTAL Base BID PLUS BID OPTION ___________

NOTES FOR BIDDING SCHEDULES

NOTE NO. 1. To better facilitate the receipt and proposal process, all modifications to proposals are to be submitted on copies of the latest bid schedules as published in the solicitation or the latest amendment thereto. In lieu of indicating additions/deductions to bid items, bidder should state the revised prices for each item. The company name should be indicated on the face of the bidding schedule to preclude being misplaced.

NOTE NO. 2. Bidder must insert a price on all numbered items of the Bidding Schedule. Failure to do so will disqualify the bid.

NOTE NO. 3. In accordance with FAR 52.211-18 Variation in Estimated Quantity -in the unit-priced line items 0004, 0005, 0006, 0012, 0013,0016, and 0018 are an estimated quantity. The actual quantity of the unit-priced item may vary. If the variance is less than 15 percent, a modification will be processed to add the quantity of the variance at the awarded unit price. If the variance is more than 15 percent above or below the estimated quantity, an equitable adjustment in the contract unit price shall be made upon demand of either party. The equitable adjustment shall be based upon any increase or decrease in costs due solely to the variation above 115 percent or below 85 percent of the estimated quantity.

NOTE NO. 4. 52.217-5 EVALUATION OF OPTIONS (JUL 1990)
Except when it is determined in accordance with FAR 17.206(b) not to be in the Government's best interests, the Government will evaluate offers for award purposes by adding the total price for all options to the total price for the basic requirement. Evaluation of options will not obligate the Government to exercise the option(s).

(End of clause)
The Government may require the delivery of the numbered line items, identified in the schedule as option items, in the quantity and at the price stated in the schedule. Subject to the availability of funds, the Contracting Officer may exercise Bid Option 1 by written notice to the Contractor within the periods expressed in the explanation of bid items.

END OF BID SCHEDULE

EXPLANATION OF BID ITEMS

GENERAL: This section comprises an explanation of the bid items identified in the bid schedule. This section is a general scope of work for the bid items described in the bidding schedule and is not intended to be all encompassing in the descriptions. All work specified herein shall be accomplished in accordance with the procedures prescribed in the technical provisions of the specifications and the Scope of Work. The Contractor shall bid each type of work under the applicable bid item. Payment described for the various bid items will be full compensation for all labor, materials, and equipment required to complete the work. Compensation for any item of work described in the contract but not listed in the bid schedule shall be included in the payment for the item of work to which it is made subsidiary.

BASE BID

Bid Item No. 1: Site Work

Payment under Bid Item No. 1, Site Work will be made at the contract job price and will constitute full compensation for all work associated with site preparation. This includes but not limited to contractor furnished office space for contractor and Government trailer, temporary utility connections, traffic control, traffic signage, temporary sanitary facilities, and project sign.

Bid Item No. 2: Marine Plant (Tugboat and Barge)

Payment under Bid Item No. 2, Marine Plant (Tug and Barge), will be provided at the contract job price and will constitute full compensation for all work associated with furnishing and operating the marine plant in accordance with the scope of work. This includes but not limited to tugboat operator, crew, initial mobilization, and final demobilization.

Bid Item No. 3: Intermediate Mobilization and Demobilization

Payment under Bid Item No. 3, Intermediate Mobilization and Demobilization will be made at the contract each price and will constitute full compensation for all work associated with intermediate demobilization and mobilization. This includes demobilizing all construction equipment from the lock chamber and remobilizing back into the lock chamber. Contractor shall enter the number of mobilizations required based upon their proposed schedule and the unit price for each mobilization and demobilization. Contractor shall be required to demobilize as directed in the specifications.

Bid Item No. 4: Horizontal 7” Core Drill for Thru-Bolt Anchor

Payment under Bid Item No. 4, Horizontal 7” Core Drill will be made at the contract linear feet price and will constitute full compensation for all work associated with drilling 7” core holes and furnishing necessary equipment to safely drill holes in vertical faces of monolith in accordance with the scope of work. Core holes for this item will extend through the monolith and daylight on the riverside of the monolith.

Bid Item No. 5: Inclined 8” Core Drill for Embedded Anchor

Payment under Bid Item No. 5, Inclined 8″ Core Drill, will be made at the contract Linear Feet price and will constitute full compensation for all work associated with drilling 8″ core holes and furnishing the necessary equipment to safely drill holes in the vertical faces of the monolith in accordance with the scope of work. Core holes for this item will extend into the monolith but will not daylight on the river side.

Bid Item No. 6: Inclined 9″ Core Drill for Toe Block Embedded Anchor

Payment under Bid Item No. 6, Inclined 9″ Core Drill, will be made at the contract Linear Feet price and will constitute full compensation for all work associated with drilling 9″ core holes and furnishing the necessary equipment to safely drill holes in the vertical faces of the monolith in accordance with the scope of work. Core holes for this item will extend into the monolith but will not daylight on the river side.

Bid Item No. 7: Monolith Monitoring

Payment under Bid Item No. 7, Monolith Monitoring, will be made at the contract job price and will constitute full compensation for all work associated with monitoring the monolith for movement during the coring process and anchor installation.

Bid Item No. 8: Anchor End Recess

Payment under Bid Item No. 8, Anchor End Recess will be made at the contract each price and will constitute full compensation for all work associated with the coring the Recess for all anchor types. This includes but not limited to providing approximately 18″ diameter recess pockets at a depth accommodating contractor's anchorage at each anchor end. The anchors must not protrude from the face of the monolith. Toe Block recess are to be formed and not included in this bid item

Bid Item No. 9: 2.5″ Diameter Anchor Install for Thru-Bolt Anchorage

Payment under Bid Item No. 9, 2.5″ Diameter Anchor Install for Thru-Bolt will be made at the contract each price and will constitute full compensation for all work associated with the anchor furnishing and installation. This includes but not limited of, providing all equipment, personnel, rods, testing/inspection, annulus grouting (calculated grout volume in core holes)and material to install anchors as shown on the contract documents.

Bid Item No. 10: 2.5″ Diameter Anchor Install for Embedded Anchorage

Payment under Bid Item No. 10, 2.5″ Diameter Anchor Install for Embedded Anchorage will be made at the contract each price and will constitute full compensation for all work associated with the anchor installation. This includes, but is not limited to, providing all equipment, personnel, rods, testing/inspection, annulus grouting (calculated bond zone and free stressing length volume)and material to install anchors as shown on the contract documents.

Bid Item No. 11: 3″ Diameter Anchor Install for Embedded Anchorage at Toe Block

Payment under Bid Item No. 11, 3″ Diameter Anchor Install for Embedded Anchorage at Toe Block will be made at the contract each price and will constitute full compensation for all work associated with the anchor installation. This includes, but is not limited to, providing all equipment, personnel, rods, testing/inspection, annulus grouting (calculated bond zone and free stressing length volume)and material to install anchors as shown on the contract documents.

Bid Item No. 12: Cementitious Grouting

Payment under Bid Item No. 12, for Cementitious Grouting will be made at the contract cubic feet price and will constitute full compensation for all work associated with pre-grouting core holes to fill and seal cracks and grouting not included in bid items No 9, 10, and 11.

Bid Item No. 13: Re-Drilling

Payment under Bid Item No. 13, for Re-Drilling will be made at the contract linear feet price and will constitute full compensation for all work associated with re-drilling after pre-grouting of cracks in the bore holes.

Bid Item No. 14: Water Stop

Payment under Bid Item No. 14 Water Stop, will be made at the contract job price and will constitute full compensation for all work associated with the water stop. This includes, but is not limited to, providing all equipment, personnel, and materials to core the bore hole, grout foundation material up to 2’ into monolith concrete, re-drill to base of core, and install the water stop as shown in the contract documents.

Bid Item No. 15: Lock Dewatering

Payment under Bid Item No. 15, Lock Dewatering, will be made at the contract job price and will constitute full compensation for all work associated with dewatering the lock. This includes but is not limited to: providing all equipment; small cranes set on the lock wall to support personnel and small lifts into/out of the lock chamber, floating crane, associated floating plant, and portable air compressor for maneuvering, handling and placing stoplogs(five stoplogs will be placed in the downstream slot and six stoplogs will be placed in the upstream slot), setting three valve culvert bulkheads (upstream of the landside upstream valve and downstream of each downstream valve), portable hydraulic dewatering pumps, erection scaffolding for ingress/egress to the lock chamber; personnel; and miscellaneous materials. After lock chamber is dewatered, the contractor will maintain dewatered status by managing the dewatering pumps, addressing leakage from the stoplogs and other water intrusions. The contractor will remove debris from the work area in the lock chamber. The Government will provide the stoplogs and picking beams, on barges, and the valve bulkheads.

Bid Item No. 16: Epoxy Grout Injection of Cracks

Payment under Bid Item No. 16, Epoxy Grout Injection of Cracks will be made at the contract gallon price and will constitute full compensation for all work associated with the surface crack grouting. This includes, but is not limited to, providing all equipment, personnel, and materials to install epoxy grout as shown in the contract documents.

Bid Item No. 17: Concrete Toe Block

Payment under Bid Item No. 17, Concrete Toe Block, will be made at the contract job price and will constitute full compensation for all work associated with the toe block installation. This includes, but is not limited to, providing all equipment, personnel, and materials to construct reinforced concrete toe block as shown in the contract documents. Toe block will be constructed underwater.

Bid Item No. 18: Polyurethane Grout Injection of Cracks

Payment under Bid Item No. 18, Polyurethane Grout Injection of Cracks will be made at the contract gallon price and will constitute full compensation for all work associated with the surface crack grouting. This includes, but is not limited to, providing all equipment, personnel, and materials to install polyurethane grout as shown in the contract documents.

Bid Item No. 19: MISC. Concrete Work

Payment under Bid Item No. 19, MISC. Concrete Work will be made at the contract job price and will constitute full compensation for all work associated with concrete repair. This includes, but is not limited to, providing all equipment, personnel, and materials to repair miscellaneous concrete as shown in the contract documents.

Bid Option

Bid Item No. 20: Remove Riverside Filling Valve

Payment under Bid Item No. 20, Remove Riverside Filling Valve, will be made at the contract job price and will constitute full compensation for all work associated with removing the riverside filling valve, trunnion beam, connecting rods, A-frame, bellcrank, and bellcrank supports. This includes but is not limited to: providing all equipment; small cranes set on the lock wall to support personnel and small lifts into/out of the valve pit; floating crane and associated floating plant for pulling and handling the valve, trunnion beam, connecting rods, A-frame, bellcrank and supports. This bid option can be exercised and awarded up to 240 days after the notice to proceed in accordance with FAR 52.217-7.

- END OF SECTION –

SECTION 01 00 00

ADDITIONAL SPECIAL CONTRACT REQUIREMENTS

PART 1 GENERAL

AMENDMENT 0002

****

1.1 SCOPE OF WORK

Holt Lock and Dam is located approximately 135 miles above the confluence of the Black Warrior and Tombigbee Rivers, and about 6 miles upstream of the City of Tuscaloosa, Alabama. On 31 May 2024, it was found the lock monolith R14 had crack along the length of the monolith. On 28 September 2024 a steel bulkhead and bulkhead frame was installed at the culvert intake to isolate the cracked monolith from upper pool allowing safe operation of the lock.

The objective of this project is for the Contractor to provide all labor, materials, and equipment necessary to repair monolith R14 in accordance with the plans and specifications. Work includes but not limited to horizontal drilling, installing horizontal post tension anchor bars, internal pressure grouting, epoxy injection of surface cracks, installing water stops, underwater concrete placement, and dewatering the lock chamber at Holt Lock.

During the construction period, the contractor shall have access to the work area from 12:00 AM Monday through 11:59 PM Friday (Central Time). The lock shall be open to the public from 12:00 AM Saturday to 11:59 PM Sunday. To facilitate commercial and public lockage, the contractor shall remove all construction equipment from the lock chamber by 11:59 PM Friday. While the lock is open to commerical traffic, there shall be no protruding anchor rods or equpmintent into the lock chamber. See drawings for work barge staging area. The Goverment expects at a minimum the contractor to work two (2) ten (10) hour shifts per work day.

To install anchor rods below the tailwater elevation, the Government will close the lock for ~~thirty (30)~~***forty-five (45)*** days to allow the contractor to dewater Holt Lock. The contractor shall have access to the work area during the dewatering 24 hours a day 7 days a week. This ~~thirty-day~~ ***forty-five (45) day*** closure period includes the installation and removal of stoplogs, dewatering the lock chamber, and refilling it back to the lower pool. The contractor is responsible for setting and removing the upper and lower stoplogs, as well as setting and removing the upper land culvert bulkhead and both downstream lower bulkheads. The contractor must dewater the lock chamber, all culverts and laterals to facilitate Government inspections and Construction Quality Assurance. Additionally, the contractor must provide access at the upper end of the lock to the dewatered lock floor, culverts, and laterals. ***After contract award the Government will coordinate with the contractor to remove silt and devris from the lock chamber in front of monolith R-14. The Government will use its floating plant to remove silt and debris at the government expense. Remaining silt and debris that the floating plant cannot be remove by the Government's floating plant will be at the contractor's expense. The Government will provide an updated hydro survey after silt and debris is removed to the contractor.*** The Goverment expects at a minimum the contractor to work two (2) twelve (12) hour shifts per work day.

Dewatering Process:

a. Place water level in chamber at upper pool position. Open the upstream miter gates. Leave the upstream valves in the open position, downstream valves in closed position, and downstream miter gates in closed position.

b. Move the floating plant (crane and upstream stoplog barge) immediately downstream of the upstream stoplog recess.

c. Install six upstream stoplogs. When finished, place the picking beam on the barge and disconnect it.

d. Place the two valve culvert bulkheads in the upstream recesses of the two upstream valves. Leave the upstream valves in the open position.

e. Use the floating plant to place a valve culvert bulkhead in the downstream recess of one of the two downstream valves.

f. Use the downstream valve that is not bulkheaded to drain the lock chamber down to the upper miter sill level.

g. Remove the 6-inch drain plugs upstream of the miter gate sill to drain the three feet of water between the stoplog sill and the miter gate sill. The plugs must be removed under water.

h. Lower the lock chamber to lower pool elevation by opening the downstream valve that is not bulkheaded. Leave the valve in the open position.

i. Open the downstream valve that is bulkheaded. Leave the valve in the open position.

j. Use the floating plant to place a valve culvert bulkhead in the downstream recess of the remaining downstream valve.

k. Open the downstream miter gates and move the floating crane and barge with the picking beam out of the lock and downstream of the downstream stoplog recess

l. Install five downstream stoplogs.

m. Ensure all culvert valves are open, and begin pumping out the lock chamber.

n. Dewatering the lock chamber will require pumps mounted on a barge below the lower lock bulkheads (stoplogs). The barge mounted dewatering pumps should have capability of dewatering the lock in approximately 24 hours. Due to configuration of the lock chamber floor, more pumps will be needed within the lock for lowering water enough to gain access to the lock culverts. Contractor should have some pumps to pump water trapped in mid-section of the lock across the lower miter sill area into the area just upstream of the lower stoplogs such that the dewatering pumps can remove from the lock chamber. In addition, the contractor may need additional pumps located within the lock laterals to pump water over the lock wall in order to allow access to the lock culver system via the lateral openings (down to 6-8" of water).

o. See O&M Manual for additional details. The O&M manual references internal dewatering pumps. They are no longer in operation and contractor shall provide portable dewater pumps to be utilized. In addition, there are no cranes on site. Contractor will be responsible for providing cranes needed for dewatering.

After construction is complete and the lock chamber is filled to upper pool the contractor shall remove temporary bulkhead and set it on Government barge.

The contractor shall submit a dewatering plan for approval at least ninety (90) days prior to dewatering. The optimum time to dewater the lock chamber is between June and August. The contractor should scheduled the dewatering in this time frame and no earlier then May 18. Please refer to drawings for further details.

During the dewatering process, the Government will be conducting work on the downstream miter gates. The contractor is required to coordinate with the Government for the lock dewatering activities. The Government will arrange its own access for the lower miter gates.

Riverside Filling Valve Removal (Bid Option)

If the contractor determines that removal of the Riverside Filling Valve is necessary to facilitate equipment installation within the culvert, the Government will proceed with executing Bid Option: Remove Riverside Filling Valve.

Components requiring removal for contractor to have full unimpeded access to culvert upstream of upper riverside valve:
1. Hydraulic Cylinder
2. Rocker Assembly
3. Rocker Support Beams
4. Upper & Lower Strut
5. Radius A-Frame
6. Valve Body

Steps for equipment removal
1. Ensure valve is in the fully closed position
2. Disconnect hydraulic cylinder from the Rocker Assembly by removing Piston Rod Pin (14/4-8)
3. Close hydraulic supply lines to cylinder, disconnect flexible lines from cylinder ports, and remove bearings (14/4-10)
4. Remove hydraulic cylinder and trunnion.
5. Disconnect Rocker Assembly from Rocker Support Beams by removing Main Rocker Pin (14/5-8) and disconnect from top strut by removing Upper Strut Pin (14/7-16).
6. Remove Rocker Assembly
7. Disconnect Rocket Support Beams by removing Support Pins (14/6-2) connecting to embedded anchorages.
8. Remove Rocker Support Beams
9. Disconnect Upper Strut from Radius A-Frame by removing Intermediate Strut Pin (14/8-6)
10. Remove Upper Strut
11. Disconnect Radius A-Frame by removing Hinge Pins (14/9-2) connecting to embedded anchorages.
12. Remove Radius A-Frame
13. Disconnect Lower Strut from Valve Body by removing Lower Strut Pin (14/7-13)
14. Remove Lower Strut

15. Disconnect Valve Body from Trunnion Beam by removing Trunnion Pins (13/3-6)
16. Remove Valve Body from recess

This list provides an outline of the sequence for component removal, not a detailed means and methods. During disassembly, the contractor is responsible for ensuring that all components are properly supported or tied off to prevent uncontrolled movement or falling once pins are removed. Some pins are secured with retainer rings or keeper bars, which must be removed prior to extraction.the contractor may resort to burning them out with a thermal lance—pending approval from the COR.

The Goverment will provide a barge for contractor to place components required to be removed from the valve pit. Installation of new vavlve and componets will be performed by others.

1.2 COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK

The Government will provide a partial notice to proceed (NTP) for the project that will allow the contractor to complete the work below. The partial NTP will provide approximately 90 days, for the start or complete these tasks prior to the issuance of a full NTP through a bilateral contract modification which will start the period of performance and establish the contract completion dates. After issuance of the full NTP, the Contractor shall be required to prosecute the work diligently, and complete the entire work ready for use not later than 270 calendar days after receipt of notice to proceed. This time stated for completion shall include final cleanup of the premises.

The proposed scope for the partial NTP is as follows:
Kick Off Meeting (This will go over contractual items and front-end information).
Site Visits
Preconstruction Submittals
Shop Drawings
Product Data
Design Data
Test Reports
Certifications
Dewatering Plan
Preliminary Schedule
Submittal and procurement of long lead items
****

AMENDMENT 0002

1.3 LIQUIDATED DAMAGES--CONSTRUCTION

(a) If the Contractor fails to complete all of the work within the time specified in the contract, the Contractor shall pay liquidated damages to the Government in the amount $1,917.00 for each calendar day of delay until the work is completed or accepted.

(b) If the Government terminates the Contractor's right to proceed, liquidated damages will continue to accrue until the work is completed. These liquidated damages are in addition to excess costs of repurchase under the Termination clause.

1.4 SECURITY REQUIREMENTS

(a) All contractor employees, to include sub-contractor employees, requiring access to Army installations, facilities shall complete AT Level I awareness training within 30 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. AT Level I awareness training is available at the following website: http://jko.jten.mil/courses/atl1/launch.html; or it can be provided by the RA ATO in presentation form which will be documented via memorandum.

(b) The contractor and all associated sub-contractors shall brief all employees on the local iWATCH, Corps Watch, or See Something, Say Something program (training standards provided by the requiring activity ATO). This local developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within 30 calendar days of contract award and within 30 calendar days of new employees commencing performance with the results reported to the COR NLT 5 calendar days after contract award.

(c) All contract employees, including subcontractor employees must obtain a T1 back ground check (standard background check).

(d) The Contractor must pre-screen Candidates using the E-verify Program (http://www.uscis.gov/e-verify) website to meet the established employment eligibility requirements. The Vendor must ensure that the Candidate has two valid forms of Government issued identification prior to enrollment to ensure the correct information is entered into the E-verify system. An initial list of verified/eligible Candidates must be provided to the COR no later than 3 business days after the initial contract award." *When contracts are with individuals, the individuals will be required to complete a Form I-9, Employment Eligibility Verification, with the designated Government representative. This Form will be provided to the Contracting Officer and shall become part of the official contract file.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Site Plan; G,CD.

Hazard Analysis; G,SO.

Construction Sequence Plan; G,CD

Dirt and Dust Control Plan; G,CD

SD-07 Certificates

Certificates of Compliance: G,CD.

SD-11 Closeout Submittals

Contractor Prepared As-Built Drawings; G,CD

1.6 CONTRACT DRAWINGS, MAPS AND SPECIFICATIONS

a) The Contractor will be furnished one CD-ROM containing a reproducible copy of the advertised solicitation, including contract clauses, plans and specifications. The work shall conform to the specifications and the contract drawings listed in the technical provisions.

b) Omissions from the drawings or specifications, the mis-description of details of work which are manifestly necessary to carry out the intent of the drawings and specifications which are customarily performed shall not relieve the Contractor from performing such omitted or mis-described details of the work but they shall be performed as if fully and correctly set forth and described in the drawings and specifications.

c) The Contractor shall check all drawings furnished him immediately upon their receipt and shall promptly notify the Contracting Officer's Representative of any discrepancies. Figures marked on drawings shall in general be followed in preference to scale measurements. Large scale drawings shall in general govern small scale drawings. The Contractor shall compare all drawings and verify the figures before laying out the work and will be responsible for any errors which might have been avoided thereby.

d)The list of drawings and maps provided in the Index Sheet of the Plans for this solicitation are hereby incorporated by reference into these specifications. Any schedules included in the drawings are for the purpose of defining requirements other than quantities.

NOTE: Refer to the folio of drawings for the index of drawings in this solicitation.

1.7 PHYSICAL DATA

Data and information furnished or referred to below is for the Contractor's information. The Government shall not be responsible for any interpretation of or conclusion drawn from the data or information by the Contractor.

a. The indications of physical conditions on the drawings and in the specifications are the result of site investigations by surveys.

b. Weather Conditions. Weather conditions shall be investigated by the Contractor to determine the hazards likely to arise therefrom. Complete weather records and reports may be obtained from the local U.S. Weather Bureau.

c. Transportation facilities. The contractor shall investigate and make arrangements for transporting all materials and equipment to the site of work.

1.8 TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER

a. This provision specifies the procedure for determination of time extensions for unusually severe weather in accordance with the contract clause entitled "Fixed Price Construction". In order for the Contracting Officer to award a time extension under this clause, the following conditions must be satisfied:

b. The weather experienced at the project site during the contract period must be found to be unusually severe, that is, more severe than the adverse weather anticipated for the project location during any given month.

1. The unusually severe weather must actually cause a delay to the completion of the project. The delay must be beyond the control and without the fault or negligence of the Contractor.

2. The following schedule of monthly anticipated adverse weather delays is based on National Oceanic and Atmospheric Administration (NOAA) or similar data for the project location and will constitute the base line for monthly weather time evaluations. The Contractor's progress schedule must reflect these anticipated adverse weather delays in all weather dependent activities.

MONTHLY ANTICIPATED ADVERSE WEATHER DELAY
WORK DAYS BASED ON (5) DAY WORK WEEK

JAN	**FEB**	**MAR**	**APR**	**MAY**	**JUN**	**JUL**	**AUG**	**SEP**	**OCT**	**NOV**	**DEC**
11	8	6	4	4	5	6	4	4	3	4	8

Upon acknowledgment of the Notice to Proceed (NTP) and continuing throughout the contract, the Contractor will record on the daily CQC report, the occurrence of adverse weather and resultant impact to normally schedule work. Actual adverse weather delay days must prevent work on critical activities for 50 percent or more of the Contractor's scheduled work day.

c. The number of actual adverse weather delay days shall include days impacted by actual adverse weather (even if adverse weather occurred in previous month), be calculated chronologically from the first to the last day of each month, and be recorded as full days. If the number of actual adverse weather delay days exceeds the number of days anticipated in paragraph 2, above, the Contracting Officer will convert any qualifying delays to calendar days, giving full consideration for equivalent fair weather work days, and issue a modification in accordance with the Contract Clause entitled "Default (Fixed Price Construction)".

1.9 TIME EXTENSIONS

Notwithstanding any other provisions of this contract, it is mutually understood that the time extensions for changes in the work will depend upon the extent, if any, by which the changes cause delay in the completion of the various elements of construction. The change order granting the time extension may provide that the contract completion date will be extended only for those specific elements so delayed and that the remaining contract completion dates for all other portions of the work will not be altered and may further provide for an equitable readjustment of liquidated damages under the new completion schedule.

1.10 CONTRACTOR PREPARED AS-BUILT DRAWINGS

a) General. In accordance with SPECIAL CONTRACT REQUIREMENT paragraph: CONTRACT DRAWINGS, MAPS AND SPECIFICATIONS, the Government will furnish the Contractor on CD ROM one electronic set of solicitation drawing files and any amendments for use in preparation of as-built drawings by the Contractor. Copies of the drawings will be the responsibility of the Contractor. The As-Built drawings shall be a record of the construction as installed and completed by the Contractor. They shall include all the information shown on the contract set of drawings and a record of all deviations, modifications, or changes from those drawings, however minor, which were incorporated in the work, all additional work not appearing on the contract drawings, and all changes which are made after final inspection of the contract work. In the event the Contractor accomplishes additional work which changes the As-Built conditions of the facility after submission of the as-built drawings, the Contractor shall furnish revised and/or additional drawings as required to depict as-built conditions. The requirements for these additional drawings will be the same as for the as-built drawings included in the original submittal.

b) Red Line As-Built Drawings. The Contractor shall have on his staff, personnel to mark up a set of paper copy construction drawings to show the As-Built conditions. These As-Built marked copies shall be kept current and available on the jobsite at all times. All changes from the contract plans which are made in the work or additional information which might be uncovered in the course of construction shall be accurately and neatly recorded, as the events occur, by means of details and notes. The Contractor shall call attention to entries by redlining areas affected. The red line As-Built will be jointly inspected for accuracy and completeness by the Contracting Officer's representative and a responsible representative of the Contractor prior to submittal of each request for payment. The Contracting Officer's Representative's approval of the current status of the as-built drawings shall be a prerequisite to the Contracting Officer's Representative's approval of request for progress payment and request for final payment under the contract. The drawings shall show the following information, but not be limited thereto:

(1) The location and description of any utility lines or other installations of any kind or description known to exist within the construction area. The location includes dimensions to permanent features.

(2) The location and dimensions of any changes within the structure.

(3) Changes in details of design or additional information obtained from working drawings specified to be prepared and/or furnished by the Contractor.

(4) All changes or modifications which result from the final inspection.

(5) Options: Where contract drawings or specifications allow options, only the option selected for construction shall be shown on the As-Built drawings.

c) Submittal of As-Built Drawings for Review and Approval: The Contractor shall participate in monthly review meetings with the Contracting Officer's Representative to show the progress made the preceding month and make all required changes. At time of final construction inspection, the

Contractor shall submit one copy of the red lined As-Built drawings to the Contracting Officer's Representative for his review and approval. The As-Built drawings shall be certified as to their correctness by the signature of an authorized representative of the Contractor. Upon Government approval of the Contractor's red-lined copy of the As-Built drawings, the Contractor shall prepare and provide one full size paper copy and four electronic sets of As-Built drawings by incorporating the red line marked up notations on the construction drawings into the electronic set of solicitation drawings and amendments. The electronic sets of the As-Built drawings shall be submitted on CD-ROM. The CD-ROM shall also contain the information contained on the project's Advertised CD-ROM including specifications and original contract drawings, Amendment, Design Analysis, and the original survey data. Submittals are to be submitted to the Contracting Officer's Representative not later than ten (10) calendar days after project completion date. The Contracting Officer's Representative will provide the full size paper set of drawings and one copy of the CD-ROM to the BCE Office. For the remaining three copies of the CD-ROM, one copy shall be kept at the Resident Office, one copy shall be provided to CESAM-EN-D (Project Architect/Engineer) and the other provided to CESAM-EN-DW (Project Support Section).

d) Final Electronic Drawing Format

(1) The solicitation drawing files and any amendments thereto will be furnished to the Contractor in electronic format. The solicitation drawing files have been prepared in Microstation or AutoCAD R13 format. The drawing file indicates the format which the drawing was developed. The Contractor shall utilize AutoCAD R14 or R2000 to revise/redraft each solicitation drawing and/or amendment drawing to reflect all changes made during construction as indicated by the red line marked up notations on the construction drawings. Revisions/redrafting shall match the font styles, sizes, and formats; line weights/thicknesses and styles/types; and all other drafting elements used on the solicitation drawing/amendments. All elements must be incorporated into each as-built drawing file; the use of reference files shall not be permitted.

(2) All revisions made to the solicitation drawings and/or amendment drawings to reflect changes made during construction shall be flagged and shall have the revision block completed as follows. The entry in the description column of the revision block shall read "AS-BUILT". The date of the revision and one approving initial from a responsible person within the Contractor's Firm shall also be included in the revision block. Above the drawing title block the drawing will be labeled in bold letters "AS-BUILT". The flagged changes and revision block format shall be in accordance with the examples shown in the Mobile District Design Manual located on the Internet at

http://www.sam.usace.army.mil/Missions/MilitaryMissions/Engineering/DesignGuides/MobileDistrictDesignGuides/MobileDistrictDesignManual.aspx

The Contractor shall also furnish a revised index of drawings to match the actual design drawings. The drawing title blocks shall be in a uniform format to match the requirements as specified in the Design Manual.

e) Payment: No separate payment will be made for the as-built drawings required under this contract, and all costs in connection there-with will be considered a subsidiary obligation of the Contract.

1.11 BULLETIN BOARD

Refer to Section 00800 for the bulletin board.

1.12 CONSTRUCTION SEQUENCE PLAN

The Contractor shall submit a Work Sequence Plan to the Contracting Officer's Representative for approval 30 days prior to commencement of construction. The Work Sequence Plan shall identify proposed sequence of performing the work identified in the scope of work.

1.13 CERTIFICATES OF COMPLIANCE

Any certificates required for demonstrating proof of compliance of materials with specification requirements shall be executed in four copies. Each certificate shall be signed by an official authorized to certify on behalf of the manufacturing company and shall contain the name and address of the Contractor, the project name and location, and the quantity and date or dates of shipment or delivery to which the certificates apply. Copies of laboratory test reports submitted with certificates shall contain the address of the testing laboratory and the date or dates of the tests to which the report applies. Certification shall not be construed as relieving the Contractor from furnishing satisfactory material, if, after tests are performed on selected samples, the material is found not to meet the specific requirements.

1.14 TEMPORARY ELECTRICAL SERVICE

All temporary electrical service on the project, and within all temporary and permanent structures shall be installed and maintained in compliance with the provisions of EM 385-1-1, latest edition, Corps of Engineers Safety and Health Requirements, and APPENDIX T of Mobile District Regulation 385-1-1, Electrical Service Requirements for Construction and Maintenance Operations. Copies of these publications are available for inspection in the District Office by Prospective bidders, and will be furnished to the successful bidder.

Electric power may be obtained within from existing sources at various locations near the work areas. Hookups to these sources shall be coordinated with and approved by the Government.

1.15 IDENTIFICATION OF EMPLOYEES

The Contractor shall be responsible for furnishing an identification badge card to each employee prior to the employee's work on-site, and for requiring each employee engaged on the work to display such identification as may be approved and directed by the Contracting Officer. All prescribed identification shall immediately be delivered to the Contracting Officer for cancellation upon the release of any employee. When required by the Contracting Officer, the Contractor shall obtain and submit fingerprints of all persons employed or to be employed on the project.

1.16 EQUIPMENT OWNERSHIP AND OPERATING EXPENSE SCHEDULE

a. This clause does not apply to terminations.

b. Allowable cost for construction and marine plant and equipment in sound workable condition owned or controlled and furnished by a contractor or subcontractor at any tier shall be based on actual cost data for each piece of equipment or groups of similar serial and series for which the Government can determine both ownership and operating costs from the contractor's accounting records. When both ownership and operating costs cannot be determined for any piece of equipment or groups of similar serial or series equipment from the contractor's accounting records, costs for that equipment shall be based upon the applicable provisions of EP 1110-1-8, "Construction Equipment Ownership and Operating Expense Schedule," Region III. Working conditions shall be considered to be average for determining equipment rates using the schedule unless specified otherwise by the contracting officer. For equipment not included in the schedule, rates for comparable pieces of equipment may be used or a rate may be developed using the formula provided in the schedule. For forward pricing, the schedule in effect at the time of negotiations shall apply. For retrospective pricing, the schedule in effect at the time the work was performed shall apply.

c. Equipment rental costs are allowable, subject to the provisions of FAR 31.105(d)(ii) and FAR 31.205-36. Rates for equipment rented from an organization under common control, lease-purchase arrangements, and sale-leaseback arrangements will be determined using the schedule, except that actual rates will be used for equipment leased from an organization under common control that has an established practice of leasing the same or similar equipment to unaffiliated lessees.

d. When actual equipment costs are proposed and the total amount of the pricing action exceeds the small purchase threshold, the contracting officer shall request the contractor to submit either certified cost or pricing data, or partial/limited data, as appropriate. The data shall be submitted on Standard Form 1411, "Contract Pricing Proposal Cover Sheet."

1.17 CONTRACTOR MAINTENANCE

At the end of each working day the Contractor shall police the work area and the area immediately surrounding the work area of all work-related debris. The Contractor shall comply with all applicable safety requirements and shall conduct his operations in a manner to insure an accident-free environment.

1.18 CONSTRUCTION/SITE MANAGEMENT STANDARDS

a. GENERAL

1) The following standards relate to the appearance of the construction site during the construction cycle, temporary administrative and storage areas, and service facilities needed for execution and completion of the work.

2) The Government will provide the Contractor with land side staging area and one river mooring area for barges. Both sites are shown on the drawings.Contractor to secure their own vessels, mooring points are not provided along the river bank. The parking area adjacent to the upper miter gates is for USACE personnel and cannot be used as a staginging area. Contractor may request temporary usage of the parking area through the COR as needed. The contractor shall submit a written request for approval.

3) Main Entrance Gate (USACE Parking AREA); Contractor shall control

access to the project site by manning the gate when the gate is in the open position and or unlocked, gate may be unmanned when the gate is in the closed position and locked. The contractor may put a double pad lock on the lower west gate. Lower west gate, Contractor shall control access to the project site by manning the gate when the gate is in the open position and or unlocked, gate may be unmanned when the gate is in the closed position and locked. The Contractor shall place a road closure sign and traffic control at the intersection of Holt Lock and Dam Road and the Resource Office building access road. The contractor shall place a road closure sign at the intersection of Holt Lock and Dam Road and USCAE maintance road (COR to identify location on site).

4) A visually acceptable site is an important construction standard. A clean, well-kept site will help ensure compliance with the safety and environmental requirements of the contract. The contractor shall maintain the trailers or storage buildings in good condition or they shall be removed. The contractor is responsible for the security of his property and general housekeeping of the area(s).

5) Site Plan. Prior to starting the work, the Contractor shall submit site plans to the Contracting Officer's representative for approval showing the layout and details of all temporary facilities used for this contract. The plan shall include the location of:

a) Safety and construction entrances
b) Trash dumpsters
c) Temporary sanitary facilities
d) Worker parking areas
e) Contractor trailer and contractor provided goverment trailer
f) Contractor storage areas
g) Staging areas
h) Temporary utility tie-ins
i) Traffic control and signage
j) Project sign

6) Site photographs prior to the start of work may be included with the plan. At completion of work, the contractor shall remove the facilities and restore the site to its original condition.

7) Dirt and Dust Control Plan: The Contractor shall submit truck and material haul routes along with a plan for controlling dirt, debris and dust on project roadways. As a minimum, the plan shall identify the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

A. CONTRACTOR'S TEMPORARY FACILITIES

1) Administrative Field Offices and Material Storage Trailers: Contractor's administrative field office and storage trailers shall be in like new condition. Storage of materiel's/debris under the trailers is prohibited.

2) Material Storage Area:

a. Staging Area - Are indicated on the drawings.

b. Temporary Sanitation Facilities. All temporary sewer and sanitation facilities shall be self contained units with both urinals and stool capabilities. Ventilate the units to control odors and fumes and empty and clean them at least once a week or more often if required by the contracting officer. The doors should be self-closing. The exterior of the unit will match the base standard. Locate the facility behind the construction fence or out of the public view.

1.19 COORDINATION CONFERENCES

Routine coordination conferences will be scheduled by the Contracting Officer's Representative throughout the life of this contract. Coordination conferences will be held to discuss contract administration, Contractor quality control, phasing, scheduling, and other aspects relating to this construction. The Using Agency, Corps of Engineers and the Contractor will be represented at each of these meetings. Similar information concerning replacement personnel shall be forwarded to the Contracting Officer's Representative, should any replacement be required at any time during the life of this contract. Coordination conferences will be scheduled to occur on a weekly basis. The Contractor shall develop the Meeting Minutes for each Coordination conference. A copy of the meeting minutes shall be provided to the Corps and all attendees via e-mail no later than three (3) working days after each meeting. The Contractor shall develop and maintain a list of action items that arise during construction or at each Coordination Conference. The Action Items list shall describe each Issue/Action Item and state what organization/person is tasked with its resolution. Blanks, or cells, shall be provided for dates when the Issue/Action Item was first raised, the due date for its resolution, and the date of actual resolution.

1.20 REGISTRATION OF PRIVATELY OWNED VEHICLES

All vehicles owned, leased, or operated on the reservation shall be registered as directed by the COR at the pre-construction conference. Evidence of required insurance must be presented at the time of registration.

1.21 CONTRACTOR FURNISHED GOVERNMENT TRAILER

The Contractor shall provide and set up for the exclusive use of the Government a new (or new condition) office trailer with a minimum of 220 square feet of usable space. The trailer shall have two office areas, of which, one area shall be large enough to hold a conference table capable of seating six (6) people. The trailer shall be complete with restroom including sink, kitchen area furnished with microwave oven and refrigerator with ice maker, and heating and air conditioning.

Furniture shall also be provided. Furniture shall include, as a minimum, two (2) desks with rolling office chairs, two (2) lockable two drawer filing cabinets, two (2) trash receptacles and conference room table and chairs capable of seating six (6) people. The trailer shall be located as directed by the Contracting Officer's representative. The trailer shall be set up within (30) calendar days of contract Notice to Proceed. Floor plan shall be approved by the Government. Setup shall include blocking, leveling, tying down; new (or nearly new) steps with covered landing at each door, and skirting. Blocking, leveling and tying down shall be in accordance with applicable state and local code requirements.

Temporary electricity, internet access, potable water, and sewer facilities shall be provided and maintained throughout the life of the contract by the Contractor. Phone service will provided by the

Government. During the life of the contract, the Contractor shall provide all maintenance and janitorial services necessary to keep the Government trailer clean and in good working condition. Weekly janitorial service shall be provided and shall include, as a minimum, removal of trash, sweeping and mopping of floors, dusting, and cleaning and deodorizing restroom. Items such as garbage bags, paper towels, toilet tissue and hand soap shall be provided. Maintenance shall include any necessary repairs to the trailer, skirting, steps or utilities and keeping the exterior of the trailer and adjacent grounds neat and clean. At the completion of the contract, the trailer and utility disconnects shall be removed by the Contractor. The trailer is the responsibility of the Contractor upon contract completion.

1.22 HAZARD ANALYSIS

A Hazard Analysis Plan, as described in Section 1, Article 01.A.05 of the Corps of Engineers Safety and Health Requirements Manual, EM 385-1-1, latest edition, is required for this contract, and shall be submitted thirty (30) days after Notice To Proceed (NTP).

1.23 ARMY PROJECT SIGN

The Contractor shall furnish and install a project sign and a safety performance sign at the location designated by the Contracting Officer within 30 days after notice to proceed. The signs shall be constructed as indicated on the figures bound herein. Size, lettering, color, and paint shall conform to the details shown in Figure 5B "Construction Sign," Figure 5C "Fabrication and Mounting Guidelines," and Figure 5D "Safety Performance Sign," bound herein. All parts of frames and signs shall be given a primer coat of oil paint and a minimum of two finish coats of white semi-gloss paint. The Contractor shall maintain the signs in a "like new" condition throughout the life of the project, repainting and replacing members as necessary to accomplish this requirement. No direct payment will be made for the signs nor maintenance of the signs.

1.24 CONSTRUCTION MATERIALS

All construction materials shall remain in the designated staging area until ready for use. Storage of materials in areas other that the designated area will not be allowed unless approved by the COR.

1.25 CONSTRUCTION DEBRIS DISPOSAL

All construction debris shall be disposed of at an approved disposal site off Government controlled lands.

1.26 RATES OF WAGES

Wage rates follow at the end of this section.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION (NOT APPLICABLE)

-- End of Section --

SECTION 13 51 00.10

INSTRUMENTATION
07/13

PART 1 GENERAL

1.1 SCOPE

The Contractor shall obtain the services of an independent firm, or present qualified personnel, specializing in the monitoring of instrumentation of the type specified herein. They shall be responsible for the work described in this section and indicated on the drawings and shall consist of the following activities:

a. Furnishing all labor;

b. Performing the required readings, in accordance with the specified schedules, on the instruments; and

c. Reporting the instrumentation data to USACE in accordance with the specified time schedule. All equipment used for instrumentation testing or observations will be furnished by the government.

The instrumentation, which includes manual measurement devices, shall include 3D crackmeters and standard crackmeters.

1.2 SYSTEM DESCRIPTION

See Appendix B.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Data Collection Form; G, ED

SD-03 Product Data

DATA; G
Submit data reports in accordance with paragraph DATA.

1.4 DAMAGE TO EXISTING PROJECT FEATURES

Locations shown for the instruments are approximate. Any damage done by the Contractor to any project features while accessing or installing instrumentation locations shall be repaired through procedures approved by the Government, by the Contractor to the original conditions or better at no additional expense to the Government. Any damage to the instrumentation and corresponding equipment is required to be reported to the Contracting Officer immediately upon discovering the damage.

PART 2 PRODUCTS

2.1 DATA

2.1.1 Data Ownership

All data generated on site by instrumentation and monitoring, and other data associated with the instrumentation of the project shall be the property of the Government.

2.1.2 Data Integrity

The Contractor is required to maintain integrity of data such that records are accurate, internally consistent, that all data and records reflect the quality of the data gathered on the site, and that all data is preserved and archived for future use.

2.1.3 Requirements for Data Plotting and Presentation

This section presents the requirements for collection of the data. The Contractor shall create a form, subject to approval, on which to record each data set. Readings shall be taken at the intervals specified in Part 3 EXECUTION. Each data set shall include but not be limited to, the time, date, Upper Pool elevation, Lower Pool elevation, lock chamber pool elevation. The name of the representative who collected the readings shall also be included, as well as the name of the representative who performed quality control on the data.

PART 3 EXECUTION

AMENDMENT 0002
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3.1 READING REQUIREMENTS

The instruments shall be read for the entirety of the contract at ~~the~~ ***all*** times ~~specified~~***while work is being performed***. The Quality Control (QC) Manager shall review monitoring data daily and with any significant activities and report daily. The Contractor shall schedule with Engineering before first anchor to assist with monitoring. This does not preclude the contractor from adding instrumentation with government approval. The Contracting Officer has the authority to request additional readings at any time. Data shall be recorded on Data Collection Form approved by the Contracting Officer. All data acquired from the monitoring program shall be the property of the Government. Contractor to coordinate monitoring with government at the preconstruction meeting with historical data and trends.

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AMENDMENT 0002

3.2 ADDITIONAL READINGS

The Contractor shall perform, upon request by the Contracting Officer, any readings in addition to the required readings.

-- End of Section --

SECTION 31 68 13

EMBEDDED ANCHORS WITH GROUTED BOND ZONES
11/20

PART 1 GENERAL

1.1 SYSTEM DESCRIPTION

Prior to commencing any work on the anchors, the Contractor, including all field personnel to be involved in drilling and installation of the anchors, must meet with the Contracting Officer to review the drawings and specifications, work plans, and submittals. Drilling may commence upon approval of the anchor installation plan and procedures described in paragraph SUBMITTALS and after the conduct of the Preparatory Meeting.

1.1.1 General Requirements

The work includes design, fabrication, and installation of embedded anchors with grouted bond zones as shown on the drawings. The overall project includes through-bolted anchors, but these specifications do not apply to them except as noted in the specification. Embedded anchors must be threaded bar type. The contractor will select and install the anchor system components to meet the minimum anchor requirements provided on the drawings and in the specifications. Prepare fabrication and installation drawings and an anchor installation work plan. Submit drawings and detailed installation procedures and sequences showing complete details of the installation procedure and equipment; anchor fabrication; grouting methods; grout mix designs; anchor placement and installation; corrosion protection for bond length, stressing length, and anchorage; anchorage and trumpet; stressing and testing procedures with lengths, forces, deformations, and elongations for approval by the Contracting Officer. Shop drawings for anchors must include, but not limited to, locations and details of the centralizers, grout tubes, and corrosion protection. If different types of anchors are to be installed, each anchor type must be readily identifiable. Once reviewed by the Contracting Officer, no changes or deviation from shop drawings will be permitted without further review by the Contracting Officer.

1.1.2 Scope of Work

The work includes the design, fabrication, and installation of the embedded anchor system. Design consists of the contractor selecting the anchor system components, determining means and methods to install them to meet the minimum anchor requirements provided on the drawings, and developing the methodology for the required testing. The intent is not for the contractor to do a full design including load derivation and anchor sizing/layout. The anchors shall be fabricated and installed as shown on the drawings and specified in the specifications. General design criteria and installation requirements are shown on the drawings. The materials, design, stressing, load testing, and acceptance must be in accordance with PTI DC35.1, the drawings, and these specifications.

a. Embedded anchors will be threaded bar type. The Contractor is responsible for the design of the anchor and bearing plate, determining drilling methods within the constraints of this specification, and determining bond zone location within each installation location shown on the drawings based on site conditions (i.e. concrete condition and geometry). Submit design computations and data for the embedded anchors, bearing plates, and bond zones.

b. Submit a plan for installing the embedded anchors for review and comment. The proposal must describe the sequence for installation and other restrictions as outlined on the drawings or specified. Determine the anchor installation procedures as part of the anchor design. Also include descriptions of methods and equipment to be used for alignment checking of anchor holes. Payment for embedded anchors must include all costs in connection with designing, fabricating, installing, and testing the anchors.

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1.1.3 Anchor Design

Design the individual Embedded anchors to meet the following criteria:

Anchor Location	As indicated on the drawings
Horizontal and Vertical Spacing	As shown on the drawings
Hole Diameter	The hole diameter will be as shown on the drawings S-401 and S-403 for each anchor location.
Design Load	As shown on the drawings
Assumed Ultimate Concrete-Grout Bond Strength	As Shown on the drawings
Minimum Unbonded Length	~~A minimum of 5 feet of unbonded anchor shall be maintained between the top of the bond zone and the last structural crack encountered during coring.~~***As shown on the drawings.***
Minimum Required Bond Length	As shown on the drawings
Maximum Anchor Hole Length	~~Concrete: A minimum of 4 feet away from monolith exterior face~~***The distal end of each anchor hole must terminate more than 4 feet away from the river side face of the monolith.***
Corrosion Protection	Class I, Encapsulated Tendon
Angle of Anchor Inclination	As shown on the drawings.
Grout	Two stage grout design with the bond zone over-grouted by 5 to 8 feet over bond-breaker. 5,000 psi strength for bond zone grout.
Anchor Bottom Depth	This will be determined based on the location of structural cracks within the concrete.

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1.1.3.1 Design Load

The Design Load must not exceed 40 percent of the ultimate strength of the high-strength steel bar. The Lock-off Load must not exceed 50 percent of the ultimate strength of the high-strength steel bar. The maximum Test Load must not exceed 60 percent of the ultimate strength of the high-strength steel bar. Design the bearing plates so that the bending stresses in the plate do not exceed the yield strength of the steel when a load equal to 95 percent of the minimum specified ultimate tensile strength of the high-strength steel bar is applied and so that the average bearing stress on the structure does not exceed 3000 psi. Fabricate and install the anchorage assembly connection to the structure in accordance with AISC 325 and ACI 318.

1.1.3.2 Design Schedule

Submit a design schedule for the anchors which includes the following:

a. Anchor number.

b. Anchor design load.

c. Type and size of tendon.

d. Minimum total anchor length.

e. Minimum bond length.

f. Minimum unbonded length.

g. Details of corrosion protection, including details of anchorage and installation.

h. Details of bearing plate, sub-bearing plate, trumpet, and anchor head, and anchor head recess.

i. Details of restressing system.

j. Two, 2-foot samples of proposed corrugated encapsulation

k. Two, 1-foot samples of bare anchor rod, one of each size.

l. Samples of centralizers, grout tubes, repair tape, and heat shrink sleeves.

m. Submit the design schedule at least 30 days prior to commencement of work on the anchors covered by the schedule.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 301	(2016) Specifications for Structural Concrete
ACI 318	(2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325	(2017) Steel Construction Manual

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A500/A500M	(2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A536	(1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings
ASTM A572/A572M	(2021; E 2021) Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel
ASTM A615/A615M	(2022) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM C33/C33M	(2023) Standard Specification for Concrete Aggregates
ASTM C109/C109M	(2021) Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or (50-mm) Cube Specimens)
ASTM D1248	(2016) Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable
ASTM D1784	(2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds
ASTM D1785	(2015; E 2018) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120
ASTM D3350	(2021) Polyethylene Plastics Pipe and Fittings Materials
ASTM D4101	(2017) Standard Classification System and Basis for Specification for Polypropylene Injection and Extrusion Materials

POST-TENSIONING INSTITUTE (PTI)

PTI DC35.1	(2014) Recommendations for Prestressed Rock and Soil Anchors
PTI TAB.1	(2006) Post-Tensioning Manual

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-1807	(2023) Drilling and Invasive Activities at Dams and Levees

1.3 DEFINITIONS

Anchor Stressing - As it pertains to this specification shall include any increases or decrease of load on the anchor, such as, but not limited to, testing, alignment load, lock-off and lift-off.

Anchored Structure - The Dam, foundation or other structure to which the anchor is to transfer force.

Demonstration Test Anchor - An anchor which is performance-tested to verify design assumptions and installation practices.

Field Trials - See Paragraph GROUT FOR PRE-GROUTING. The demonstration of the ability to mix and pump three specific grout mixes to the pressures described herein while simultaneously obtaining required pressure and volumetric data.

Pilot Holes - A smaller hole drilled into a material for accuracy or convenience prior to enlarging the hole to the specified diameter. For this contract all pilot holes will be drilled using directional drilling methods.

Enlargement Drilling - Drilling to increase the diameter of an existing hole.

Artesian Conditions - Condition in which water was observed to be flowing from a borehole or open fracture at a pressure head elevation greater than the elevation of the ground surface.

Artesian Flows - Water flowing out of boreholes where "artesian conditions" are observed.

Exclusion Zone - Regarding any reference to excluding activities within a certain radius of a borehole or anchor, that zone shall consist of a cylinder around the borehole or anchor with a specified radius.

Backfill - soil or aggregate that does not contain any particles greater than 4 inches in diameter and contains a least 25 percent sand, silt, and/or clay particles by weight. Backfill must not contain roots, trash, or hazardous or toxic materials.

1.4 SITE CONDITIONS

A crack investigation of monolith 14R was conducted at the site by the government in 2024 and findings are presented in the contract drawings. Essentially, a vertical crack has split the monolith in two halves starting at the culvert and rising up through the lock control station basement. The degree of cracking beneath the culvert is not fully understood at this time and may have multiple structural cracks in those lower lifts. Variations of the monolith condition which may be encountered include, but are not limited to, zones throughout the monolith with different concrete strengths, structural cracks, embedded utilities, embedded reinforcement, pre-formed voids, leakage from the river pool and lock chamber, and other discontinuities in the concrete structure. Such variations will not be considered as differing materially within the purview of the contract clauses, paragraph differing site conditions. Foundation data are available for inspection as specified in the special contract requirements, paragraph Physical Data, but no drilling will be done into the foundation for this contract.

During this contract period, the lock will be closed to industry on the weekdays and available for the contractor to work unimpeded. The intent is to minimize effects on industry. During the weekend, the lock will be open to industry and the contractor will have limited access to the lock chamber. Therefore, anchor installation will occur with the chamber filled (or partially filled) to permit work on weekdays without setting and removing stoplogs. During work days, the lock operator is capable of raising or lowering the chamber pool level as necessary for the anchor work to progress. Since the monolith's vertical crack is in communication with the chamber, the water elevation in the crack is at the same elevation as the chamber. With lower pool at elevation 122.9 ft and given the shallow installation angles, the upper eleven rows of anchors should be able to be drilled without water intrusion into the borehole from the crack. Anchor Row 12 and 13 are at elevations 121.52 feet and 101.5 feet, respectively, which are below lower pool and will be submerged until stoplogs are set and the chamber is unwatered. Both rows of anchors are embedded anchors with grouted bond zones, no through-bolted anchors. Because Anchor Row 13 is below the wall culvert, there is a possibility that monolith cracking allows water from the foundation into the borehole. Investigations to this point cannot confirm the locations of the cracking.

1.5 UNIT PRICES

1.5.1 Drilling Holes in Concrete

1.5.1.1 Payment

Payment will be made for costs associated with Core Drilling Horizontal and Inclined Holes in concrete as shown in the plans.

1.5.1.2 Measurement

Drilling Holes in concrete will be measured for payment to the nearest foot, based upon the linear feet of hole actually drilled in concrete in accordance with the specifications.

1.5.1.3 Unit of Measure

Linear foot.

1.5.2 Pre-Grouting Holes

1.5.2.1 Payment

Payment will be made for costs associated with Pre-Grouting Holes for crack remediation.

1.5.2.2 Measurement

Pre-Grouting Holes will be measured for payment based upon the cubic feet of cement grout or neat cement grout that were actually placed into the anchor holes as specified.

1.5.2.3 Unit of Measure

Cubic Feet.

1.5.3 Re-drilling Grouted Holes

1.5.3.1 Payment

Payment will be made for costs associated with Re-drilling Grouted Holes.

1.5.3.2 Measurement

Re-drilling Grouted Holes will be measured for payment to the nearest foot, based upon the linear feet of hole actually drilled in grout in accordance with the specifications.

1.5.3.3 Unit of Measure

Linear foot.

1.5.4 Embedded Anchors, Complete

1.5.4.1 Payment

Payment will be made for costs associated with furnishing and installing embedded Anchors, Complete which are accepted. The price must include installation of anchors and proof testing as specified. No payment will be made for anchors which do not meet the acceptance criteria, except when failure is due to information furnished by the Government.

1.5.4.2 Measurement

Embedded Anchors, Complete will be measured based upon the number of anchors installed and accepted in accordance with the specifications.

1.5.4.3 Unit of Measure

Each.

1.5.5 Proof Tests

1.5.5.1 Payment

Payment will be made for costs associated with performing Proof Tests on anchors which are accepted.

1.5.5.2 Measurement

Proof Tests will be measured based upon the number of tests performed on anchors which are accepted in accordance with the specifications.

1.5.5.3 Unit of Measure

Each.

1.5.6 Performance Tests

1.5.6.1 Payment

Payment will be made for costs associated with performing Performance Tests on anchors which are accepted.

1.5.6.2 Measurement

Performance Tests will be measured based upon the number of tests performed.

1.5.6.3 Unit of Measure

Each.

1.5.7 Creep Tests

1.5.7.1 Payment

Payment will be made for costs associated with performing Creep Tests on anchors which are accepted. No payment will be made for creep tests on anchors which do not meet the acceptance criteria.

1.5.7.2 Measurement

Performance Tests will be measured based upon the number of tests performed on anchors which are accepted in accordance with the specifications.

1.5.7.3 Unit of Measure

Each.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only or as otherwise designated. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Adequate time (a minimum of 30 calendar days exclusive of mailing time) shall be allowed and shown on the register for review and approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

1.6.1 SD-01 Preconstruction Submittals

Designer Qualifications; G, DO
Fabricator Qualifications; G, DO
Installer Qualifications; G, DO
Directional Driller Qualifications; G, DO
Anchor Hole Driller Qualifications; G, DO
Hole Alignment Survey Specialist Qualifications; G, DO
Core Logging Qualifications; G, ED, DO

Design Schedule; G, ED, DO

Anchor Installation Work Plan; G, ED, DO

Anchor Installation Schedule; G,DO
Anchor Installation Construction Platforms Maintenance and Inspection Plan

1.6.2 SD-02 Shop Drawings

Fabrication and Installation Drawings; G

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1.6.3 SD-03 Product Data

Equipment; G, DO
Anchor Hardware; G, ED
Corrosion Protection; G, ED
Centralizers; G, ED
Manufactured Units; G, ED
Anchor Grout for Tendons; G, ED
Non-shrink Grout; G, ED
Drilling Equipment; G
Grout Mixer; G
Alignment Checking Equipment; G
Stressing/Testing Equipment***; G***
Plugs; G, ED
Grouting Equipment; G, DO
Cementitious Crystalline Waterproofing Additive; G, ED
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1.6.4 SD-05 Design Data

Design Computations; G
Anchor Installation Construction Platforms
Anchor Design; G

1.6.5 SD-06 Test Reports

Borehole Alignment Surveys
High-Strength Steel Bar
Grout Requirements
Pre-Grout Mixture Proportions
Anchor Grout Mixture Proportions
Grout Test Reports
Survey Evaluation Report; G
Field Quality Control; G, ED
Driller Logs; G, ED
Grouting Records; G, ED
Installation Records; G, ED
Stressing Records; G, ED
Stressing Equipment Calibration Records; G
Re-Stressing

1.6.6 SD-07 Certificates

Pull Test Jack Certification; G
Certification of Prestressing Technicians; G

Pressure Gage Calibration; G
Torque Wrench Calibration; G

Certificates of compliance stating that the materials conform to the specified requirements:

Nonshrink Grout
High-Strength Steel Bar
Epoxy-Coated Steel Bars
Cement
Bearing Plate
Epoxy Coating
Corrosion Inhibiting Compound

1.6.7 SD-09 Manufacturer's Field Reports

Pull Tests Records; G, DO

1.6.8 SD-11 Closeout Submittals

Driller Logs
Anchor Records

1.7 QUALIFICATIONS

Submit qualifications and experience records of the anchor designer, fabricator, installer, directional driller (if applicable), hole alignment survey specialist, and core logger for approval at least 45 calendar days prior to conducting any work in accordance with paragraph SUBMITTALS. The submittals must, where applicable, identify individuals who will be working on this contract. Their experience records shall identify all the individuals responsible for the anchor installation and shall include a listing of projects of similar scope performed within the specified period along with points of contact. No changes shall be made in approved personnel without prior approval of the Contracting Officer.

1.7.1 Designer Qualifications

Design consists of the contractor selecting the anchor system components, determining means and methods to install them to meet the minimum anchor requirements provided on the drawings, and developing the methodology for the required testing. The intent is not for the contractor to do a full design including load derivation and anchor sizing/layout. The anchors shall be fabricated and installed as shown on the drawings and specified in the specifications. General design criteria and installation requirements are shown on the drawings. Final design by Professional Engineers who have designed a minimum of three embedded anchor projects similar in size and scope to this project within the past ten years and shall have provided engineering support to the projects as needed from the start of construction until its completion. The drawings and calculations must be signed by the Professional Engineer.

1.7.2 Fabricator Qualifications

The anchors must be fabricated by a manufacturer that has been in the practice of designing and fabricating embedded anchors similar in size and scope to this project for at least five years.

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1.7.3 Installer Qualifications

The embedded anchors shall be installed by a firm which is regularly engaged in the installation of embedded anchors and should have at least eight years' experience in the installation of similar anchors. The firm shall have a field superintendent and a field engineer. The field superintendent shall have supervised the complete drilling, installation, stressing, and testing of anchors on at least five projects of similar scope and size. The field engineer ~~or geologist~~ shall be a Professional Engineer ~~or Professional Geologist registered in the State of Alabama,~~ with at least five years' experience with drilling, installing, stressing, and testing embedded high strength steel bar anchors of similar scope and size. In the experience record, identify all the individuals responsible for the anchors and must include a listing of projects of similar scope performed within the specified period along with points of contact. . Submit certification of prestressing technicians.

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1.7.4 Directional Driller Qualifications

The directional drilling shall be performed by a firm which has at least two (2) projects showing experience directionally drilling through concrete on a 1:200 tolerance as measured against a specified center line with real time survey and directional correction capabilities. The driller shall have completely and successfully drilled holes using directional drilling techniques on holes at angles as high as 15 degrees from horizontal, and at minimum tolerances of 1 in 200, or greater, on at least two projects.

1.7.5 Anchor Hole Driller Qualifications

Submit the qualifications and experience records for approval. In the experience record, identify all the individuals responsible for anchor hole drilling and must include a listing of projects of similar scope performed within the specified period along with points of contact. Qualifications prior to the drilling of any anchors specified in this section. The holes must be drilled by a firm which is regularly engaged in the drilling of embedded anchors and has at least ten years experience in the drilling of similar anchor holes. The superintendent must have drilled anchor holes on at least five projects of similar scope and size.

1.7.6 Hole Alignment Survey Specialist Qualifications

The hole alignment survey specialist shall have at least three years' experience operating the approved method being used by the Contractor to survey the alignment of the anchor holes and pilot holes. These qualifications shall be supplemented by onsite training and oversight by personnel from the manufacturer of the selected and approved alignment surveying equipment during the surveying of the first two anchor boreholes (per equipment type). If more than one survey specialist is approved the supplemental onsite training shall be repeated for the first two anchor boreholes (per equipment type) each survey specialist performs. Subsequent manufacturer oversight may be required by the Contracting Officer, if the results of surveying are suspected to be in error, inaccurate, or become otherwise questionable.

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1.7.7 Core Logging Qualifications

Logging of core and preparation of drilling logs and records must be performed by a Registered~~ Geologist or~~ Geotechnical Engineer who has at least five years experience in identifying and logging rock and concrete core. Logging the condition and characteristics of the concrete shall be part of the core logging for each cored hole.

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1.8 PREPARATORY MEETING

Prior to commencing each stage of work (e.g. platform fabrication, anchor hole drilling, ~~watertightness testing, ~~pre-grouting, ~~corrugated~~***anchor*** installation, anchor testing) on the anchors, the Contractor, including all field personnel and designers to be involved in the work, shall participate in a Preparatory Meeting with representatives of the Contracting Officer to review the plans and specifications, work plans, and submittals. Work may commence upon approval of all submittals for accessing, drilling, fabricating, installing, and testing the anchors described in Paragraph SUBMITTALS and Paragraph QUALIFICATIONS, and after the Preparatory Meeting.

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1.9 ANCHOR INSTALLATION WORK PLAN

The Contractor shall develop and submit for approval a plan for the installation and grouting of the anchors at least 45 calendar days prior to conducting any work. The initial plan shall assume that the anchors shown on the drawings will be installed. The plan shall describe the sequence for installation and other restrictions as outlined on the drawings or specified. The anchor installation procedures shall be determined by the Contractor as part of the anchor design. The installation plan shall also include grouting procedures for the corrugated tubing, bond zone, unbonded length, trumpet, method for inspecting and repairing bar corrosion protection during installation, and descriptions of methods and equipment to be used by the Contractor for alignment-checking of anchor holes during drilling and after each hole is complete. Pre-grouting sequence and mix design shall be included. Any temporary excavations associated with anchor installation must be designed and delineated in this plan. The plan shall include a specific section that addresses the equipment and methods pertaining to grouting in addition to that required for anchor installation. The stressing sequence is as shown on the construction plans, but may be altered with prior approval. The Contractor may submit a revision to the plan at least 14 calendar days prior to production installations based on documented pre-installation testing described herein and any proposed procedural refinements. In addition, the plan should include relevant requirements as outlined in ER 1110-1-1807. A hydraulic fracturing analysis will be included with the plan using the USACE Risk Management Center's Hydraulic Fracture Toolbox as documentation of the analysis. It is located at the following URL:
https://www.rmc.usace.army.mil/Software/RMC-Toolboxes/RMC-Hydraulic-Fracture-Toolbox/

The plan must include the preceeding requirements and include, but not be limited to, the following:

a. Equipment description. This includes equipment size and weight, rationale for selection, recent records of calibration, suitability to site conditions and any special considerations to be addressed due to limited site access, limited water access, and work space.

b. Description of site layout for performing the work. Provide diagrams showing needed clearance, available width of working area. Plan, profile, and cross-sections showing all pertinent project features and proposed drilling locations. Existing data, including drawings, depicting the current understanding of lock monolith conditions as they pertain to the proposed work.

c. Project personnel, including qualifications/experience

d. Description of drilling methods and techniques. Complete description of the installation procedures and materials to be used for drilling holes, diameter and length of holes, and means for inserting anchors. Include any different techniques anticipated as material types being drilled change. Include anticipated and potential adverse issues or risks posed to the lock integrity due to the proposed work. Mitigation alternatives or measures to monitor for and reduce those identified risks, including hydraulic fracture calculations. Emergency response plan if adverse impacts occur including repair alternatives, materials to be available on site, or actions that will be implemented if issues arise.

e. State the method to be used to determine that cleanout has been successful and the allowable time between final cleanout and the start of grout placement.

f. Means of mixing and placing grout, means for preventing loss of grout from holes during mixing, and means for holding the tendon in place until grout sets.

g. Proposed grout design, including but not limited to the following:
   1. Manufacturer’s certified test results of set time, shelf life, and compressive strength.
   2. Type of portland cement.
   3. Aggregate source and gradation.
   4. Proportions of mix by weight and water-cement ratio.
   5. Manufacturer, brand name and technical literature for proposed admixtures.
   6. Results of compressive strength tests performed according to ASTM C109 and completed no more than one year before the start of grouting. Use an accredited independent testing lab to verify the specified minimum 3 and 28-day grout compressive strengths.

h. Storage and handling of materials, including how materials will be maintained in compliance with manufacturer's specifications.

i. The work areas and logistics will vary across the project site, so the Work Plan shall include separate sections which clearly distinguish work areas.

j. A description of the methods used to protect materials from corrosion after delivery and prior to placement.

k. A sample Installation Records submittal containing field forms and data sheet samples that will be used for the regular submittal of installation information.

l. Grouting equipment and procedures to provide the information required in Paragraph GROUTING ANCHOR HOLES.

m. Certification page with signature block for the Mobile District Dam Safety Officer. Must contain the following statement: "This Drilling Program Plan has been developed and reviewed by experienced professionals and complies with all the requirements of ER 1110-1-1807. The proposed actions are justified and have been developed to minimize the likelihood of damage to the existing structure."

1.10 ANCHOR INSTALLATION SCHEDULE

The installation schedule shall be submitted at least 15 calendar days prior to commencement of work on the anchors covered by the schedule. The Contractor shall furnish an installation schedule for the anchors which includes the following:

. Anchor number
. Heat or lot numbers of the bar used to manufacture the anchor
. Bar creep results from each heat or lot
. Anchor Lock-off Load
. Type and size of tendon
. Total anchor length
. Bond length (1st stage grout length over the bar)
. Tendon bond length (bare bar length)
. Unbonded length
. Details of corrosion protection, including details of anchorage and installation
. Details of bearing plate, sub-bearing plate, trumpet, anchor head, anchor head recess, and where applicable, columns under the bearing plate.

1.11 DELIVERY, STORAGE, AND HANDLING

Materials must be suitably wrapped, packaged or covered at the factory or shop to prevent being affected by dirt, water, oil, grease, and rust. Protect materials against abrasion or damage during shipment and handling. Place materials stored at the site above ground on a well-supported platform and covered with plastic or other approved material. Protect materials from adjacent construction operations. Grounding of welding leads to prestressing steel will not be permitted. Reject and remove from the site prestressing steel which is damaged by abrasion, cuts, nicks, heavy corrosions, pitting, excessive heat, welds or weld spatter. Inspect tendons prior to insertion into anchor holes for damage to corrosion protection. Repair any such damage in a manner recommended by the tendon manufacturer and approved by the Contracting Officer. Lifting of pre-grouted anchors must be to manufacturers' recommendations and not cause excessive bending, which can deform the rod.

1.12 SHOP DRAWINGS

Submit for approval, at least 45 calendar days prior to conducting any work. The design shall include: drawings and detailed installation procedures and sequences showing complete details of the installation procedure and equipment; all proposed drilling methods and procedures;

water level measurement methods and procedures; hole alignment surveying methodologies; borehole airlift methods and procedures; anchor fabrication; grouting methods; grout mix designs; anchor placement and installation; corrosion protection for bond length, stressing length and anchorage; and, stressing and testing procedures with lengths, forces, deformations, and elongations for approval by the Contracting Officer. Shop drawings for anchors shall include but not be limited to locations and details of the centralizers, grout tubes, and corrosion protection. If different types of anchors are to be installed, each anchor type shall be detailed on the drawings and readily identifiable. This submittal must also describe planned procedures for any excavation and backfilling required as part of anchor installation. Once reviewed by the Contracting Officer, no changes or deviation from shop drawings shall be permitted without further review by the Contracting Officer.

1.13 DESIGN COMPUTATIONS

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1.13.1 Anchor Installation Calculations

The Contractor shall submit design computations and data for the various size anchors, including design computations and data for the bearing plate, sub-bearing plates, trumpets, anchor heads, anchor recesses and where applicable, columns under the bearing plate. The computations shall include drawings, design assumptions, calculations, and other information in sufficient detail to verify the design proposed. The design shall be certified by a Professional Engineer ~~registered in the State of Alabama~~ with proven experience in design of rock and concrete anchor systems as stated in Paragraph QUALIFICATIONS. Calculations shall be included for the stressing frames. The Contracting Officer will approve the Contractor's design calculations. Approval of the Contractor's design calculations will not relieve the Contractor of responsibility for unsatisfactory performance of the bearing plates. All design computations shall be furnished at least 30 calendar days prior to the proposed commencement of drilling.

1.13.2 Anchor Installation Construction Platforms

The Contractor shall provide design computations and drawings prepared and stamped by a Professional Engineer ~~registered in the State of Alabama~~ for any platforms required to install the anchors. The Contractor assumes all liability for the design and use of all anchor platforms. All design evaluations and calculations shall be furnished at least 30 calendar days prior to commencing construction or use of any platforms. Design of any new anchor installation construction platforms shall include and address, as a minimum the following:

a. The design shall accommodate the installation of the anchors as indicated in the drawings and specifications.

b. The Contractor shall design the platforms of sufficient width to accommodate a 5-foot-wide safe walkway behind all equipment required for the installation of the anchors.

c. The Contractor shall design legs and other appurtenances of the platform as non-adjustable. In the design of the anchorage for all platforms to the dam concrete, the Contractor shall neglect as a minimum the outer 6 inches of concrete.

d. Fracture critical members or member components used in the design of the platforms shall meet the guidance of AASHTO/AWS D1.5M, Bridge Welding Code.

e. The Contractor shall design the platforms for all impact and vibratory loading.

f. All platforms and their components shall be capable of supporting without failure at least four times the maximum anticipated load including impact and vibratory loading.

g. The Contractor shall design the platforms in accordance with applicable requirements or EM 385-1-1.

h. The Contractor shall design the platforms to be removable. All anchorages used in the design shall be completely removed from concrete and the concrete repaired as directed by the Contracting Officer.

i. When designing the platforms, the Contractor shall consider both design criteria for platforms and for bridges and shall use the more stringent of the two.

1.13.3 Anchor Installation Construction Platforms Maintenance and Inspection Plan

The Contractor shall submit an Anchor Installation Construction Platform Maintenance and Inspection Plan for any anchor installation construction platforms to the Contracting Officer for review and comment. The plan shall be prepared and stamped by a Professional Engineer ~~registered in the State of Alabama~~ who is the designer of record for the platform. The designer of record for the platform is the engineer who designed any new platforms. The plan shall, at a minimum, highlight the critical areas for maintenance and inspection of items such as critical members, connections and attachments to the lock or barge. The plan shall involve an initial inspection prior to platform use and intermediate inspections, at a minimum, of every six months for all platforms. The plan shall include a schedule for all maintenance items and all inspections. The schedule shall account for favorable weather conditions for the inspection. The Contracting Officer shall be notified for each inspection two weeks prior to any inspection of the platform. Any change to the plan or the schedule shall be reported to the Contracting Officer.

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PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 High-Strength Steel Bar

Submit certified test reports for each heat or lot of high-strength steel bar with materials delivered to the site. Submit 5 copies of mill reports and 5 copies of a certificate from the manufacturer stating chemical properties, ultimate strengths, yield strengths, modulus of elasticity, and any other physical properties needed for the required computations, for the type of steel furnished. Tension test specimens shall be the full section of the bar as rolled. Machined-reduced section test specimens are not permitted. Certified test reports will represent as ordered and as delivered condition, Type II fully threaded bar. Secondary processing is any process performed on the bar after initial testing. Secondary processing that can affect mechanical properties of the bars, shall require the bars be retested for all specified mechanical properties and performance requirements affected by the processing. This processing includes heat treatment, hot dip galvanizing, cold rolled threading and other processing that could affect mechanical properties. The supplier shall provide all certified test reports, initial testing and any test performed due to secondary processing. Relaxation testing shall be performed by an independent ISO certified lab.

2.1.1.1 High-Strength Steel Bars

Quenched and tempered fully threaded high strength bar meeting all the requirements (including supplementary) of ASTM A722/A722M Type II bar excluding the manufacturing requirements in section 4.2. Bar shall have relaxation losses of not more than 2.5% when initially loaded to 50% of specified minimum breaking strength after 1000 hours of testing in accordance with ASTM A421/A421M. Minimum 1 relaxation per heat of bar. This high strength steel will be referred to as A722-like.

2.1.1.2 Epoxy-Coated Steel Bars

Submit written certification for coating material and coated bars with the delivery of the bars. Epoxy coated steel bars must conform to PTI DC35.1 -14 Section 4.2.4. Coating at the anchorage end may be omitted over the length provided for threading the nut against the bearing plate.

2.1.1.3 Steel Rebar

ASTM A615/A615M, Grade 60.

2.1.2 Structural Steel

Bearing plate material shall be ASTM A36/A36M or ASTM A572/A572M, Grade 50.

2.1.3 Steel Pipe

ASTM A53/A53M, Type E or S, Grade B.

2.1.4 Steel Tube

ASTM A500/A500M.

2.1.5 Ductile Iron Castings

ASTM A536.

2.1.6 Polyethylene Tubing

2.1.6.1 Smooth Polyethylene Tubing

ASTM D3350 or ASTM D1248, Type III.

2.1.6.2 Corrugated Polyethylene Tubing

Minimum wall thickness of 0.06 inch as measured in any location. No recycled material shall be used in the production of the corrugated tubing.

2.1.7 Smooth Polypropylene Tubing

ASTM D4101, designation PP 210 B5542-11.

2.1.8 Polyvinyl Chloride (PVC) Pipe

ASTM D1785.

2.1.9 Polyvinyl Chloride (PVC) Tubing

2.1.9.1 Smooth Polyvinyl Chloride (PVC) Tubing

ASTM D1784.

2.1.9.2 Corrugated Polyvinyl Chloride (PVC) Tubing

Manufactured from rigid PVC compounds conforming to ASTM D1784, Class 13464-8 with minimum wall thickness of 0.04 inch as measured in any location.

2.1.10 Heat Shrinkable Sleeve

Radiation crosslinked polyolefin tube internally coated with an adhesive sealant and conforming to PTI DC35.1-14 Section 4.8.1.

2.1.11 Corrosion Inhibiting Compound

The corrosion inhibiting compound must conform to the requirements of Section 4.6 of PTI DC35.1-14.

2.1.12 Epoxy Coating

The epoxy coating required shall conform to the requirements of ASTM A775/A775M, with a minimum thickness of 0.016-inch as measured in any location.

2.2 MANUFACTURED UNITS

2.2.1 Anchor Hardware

Anchor hardware shall consist of steel bearing plate, nut, washer, beveled/wedge washer, trumpet, and any hardware recommended by the anchor manufacturer (as required), for bar anchors and corrosion protection. Submit bearing plate material and details.

Anchorage devices must be capable of developing 95 percent of the guaranteed ultimate strength of the high-strength steel bar. The anchorage devices must conform to the static strength requirements of Section 3.1.6 (1) and Section 3.1.8 (1) and (2) of PTI TAB.1. Design threaded anchorage items for epoxy coated bars to fit over the epoxy coating and maintain the capacity of the prestressing steel. Fabricate the trumpet used to provide a transition from the anchorage to the unbonded length corrosion protection from steel pipe or steel tube. The minimum wall thickness must be 0.125 inch for diameters up to 4 inches and 0.20 inch for larger diameters. Weld the trumpet to the bearing plate.

2.2.2 High-Strength Steel Bar Couplers

High-Strength Steel Bar couplers must be capable of developing 100 percent of the minimum specified ultimate tensile strength of the high-strength steel bar.

2.2.3 Centralizers

Fabricate centralizers from plastic, steel or other approved material which is nondetrimental to the high-strength steel bar. Do not use wood. The centralizer must be able to support the tendon in the drill hole and position the tendon so a minimum of 0.5 inch of grout cover is provided. Centralizers must permit grout to freely flow up the drill hole.

2.2.4 Steel Sleeve Inside Toe Block

The contractor may opt to cast steel pipe sleeves through the thickness of the new concrete toe block to aid in drilling alignment for toe block anchors. The steel sleeve running through the new concrete toe block, as shown on the drawings, must be steel pipe or steel tube selected and sized by the Contractor. Sleeve must be the necessary type and size to permit proper drilling of anchor holes into the monolith wall and placing of anchors as specified herein and shown on the drawings.

2.2.5 Anchorage End Caps

Fabricate anchorage end caps from steel. The material used must not be subject to attack by cement, wax, corrosion-inhibiting greases, or the environment. Securely attach the end cap to the bearing plate as shown on drawings. The cover shall be filled with a corrosion protection substance such as grease or wax, and the end cap must form a permanent watertight enclosure for the anchorage device.

2.2.6 Plugs

Plugs must be used to prevent riverwater from entering open drilled boreholes with no anchor installation during lock usage. Plugs shall not deface or cause damage to the monolith concrete. Contractor provides means and methods for fabrication and installation as needed.

2.2.7 Core Boxes

Use longitudinally partitioned, hinged top, core boxes constructed of approved materials in general accordance with industry state of practice to store concrete cores containing structural cracking (not mechanical breaks) in all anchors as shown on the drawings. Use as many core boxes as required to box only these sections. Furnish core boxes completely equipped with all necessary partitions. Maximum length shall be 5 feet.

2.2.7.1 Core Box Labels

At a minimum, label core boxes with the following information:

- -Project
- -Anchor Number
- -Box Number and total number of boxes for the anchor
- -Core size
- -Dates of drilling
- -Top and Bottom depths and elevations

2.3 EQUIPMENT

The Contractor's Quality Control manager shall verify that the equipment used on site is the same as the equipment submitted for approval as part of the anchor installation work plan.

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2.3.1 Drilling Equipment

Drilling equipment and methods shall be suitable for advancing each hole to the required depth, diameter, and alignment at the correct location. The Contractor may use directional or traditional equipment. The drilling systems shall be capable of drilling to ~~a minimum of 1 in 200 tolerance as~~ ***tolerances shown in drawing S-131 and*** measured against the specified anchor centerline, unless otherwise specified in this section. The equipment must be capable of core drilling a smaller diameter pilot hole and then enlarging the pilot hole to the approved anchor hole diameter. The method used for enlarging the pilot hole must follow the pilot hole and not be percussive. Enlarging the pilot hole must not cause any damage to the structure or reduce bond development between the grout and concrete.

The drilling methods shall use only water or air as a drilling fluid and be non-percussive. The Contractor shall not use drilling muds. If at any phase of constructing the anchor hole (e.g. drilling the pilot hole, enlarging the pilot hole, redrilling pre-grouted anchor holes, roughening the borehole walls, etc.), the anchor hole no longer meets the alignment tolerances, the Contracting Officer can direct the Contractor to grout the hole up and start over at no cost to the government.

2.3.1.1 Standard Drilling Equipment for Pilot or Anchor Holes

Drilling equipment and methods shall be suitable for drilling each hole to the required depth and alignment, at the correct location. Final hole diameter will vary depending on the anchor location as shown on the drawings, but shall be either 7-inch or 8-inch diameter. The drilling method shall use only water and/or air as a drilling fluid and produce core samples for borehole logging. The Contractor shall not use drilling muds. The drilling system shall be capable of drilling to ~~a minimum of 1 in 200 tolerance~~***the tolerance shown in drawing S-131***, as measured against the specified anchor centerline. Subject to the approval of the Contracting Officer, the Contractor may select the drilling means and methods provided they do not cause any damage to the structure or adversely affect the beneficial impact of any pre-grouting operation, or in any way inhibit or reduce bond development with the anchor tendon grout. The selected equipment shall meet all other requirements listed in these specifications.

2.3.1.2 Directional Drilling Equipment for Pilot or Anchor Holes

Pilot and anchor holes may be drilled using directional drilling methods provided they can provide core samples of the monolith. Directional drilling equipment and methods shall be suitable for drilling each hole to the required depth and alignment, at the correct location. Final hole diameter will vary depending on the anchor location as shown on the drawings, but shall be either 7-inch or 8-inch diameter. The directional drilling method shall use only water and/or air as a drilling fluid. The Contractor shall not use drilling muds. The directional drilling system shall be capable of drilling to ~~a minimum of 1 in 200 tolerance~~***the tolerances shown on drawing S-131***, as measured against the specified anchor centerline, and shall have real time survey and directional correction capabilities. If at any time the real time survey capabilities are lost the Contractor shall stop drilling and notify the Contracting Officer.

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2.3.1.3 Enlargement Drilling Equipment

Enlargement drilling equipment and methods, if employed, shall be suitable for enlarging each hole to the required diameter, depth, and alignment, at the correct location and with the hole being clean of cuttings. The drilling system shall be capable of following the noncompulsory pilot hole, not damage the structure, and not be detrimental to the grout-to-concrete bond. The drilling method shall use only water and/or air as a drilling fluid. The Contractor shall not use drilling muds. Upon approval of the alignment survey of a hole at the required diameter by the Contracting Officer, that hole can be prepared for anchor installation if pre-grouting has already been conducted.

2.3.1.4 Redrilling Equipment

Redrilling equipment and methods shall be suitable for redrilling each hole to the required diameter, depth and alignment, at the correct location, and with the hole being clean of cuttings. The Contractor shall use non-percussive rotary drilling equipment with no percussive methods. The drilling system shall be capable of following the original anchor borehole without changing the alignment, damaging the structure, and shall not be detrimental to the pre-grout filling the voids outside the original borehole. The drilling method shall use only water and/or air as a drilling fluid. The Contractor shall not use drilling muds.

2.3.2 Grouting Equipment

2.3.2.1 Grout Mixer

The grout mixer shall be a high-speed, high-shear, colloidal type grout mixer capable of continuous mechanical mixing that will produce uniform and thoroughly mixed grout which is free of lumps and undispersed cement. The mixer shall be equipped with suitable water and admixture measuring devices calibrated to read in gallons and tenths and so designed that after each delivery the gauge can be conveniently set back to zero. A paddle storage tank shall be used to agitate the mix prior to pumping.

2.3.2.2 Grout Pump

The grout pump shall be of the progressive cavity type, and shall be capable of pumping at rates below 20 gallons per minute, shall be capable of pumping at the pressure of at least 50 psi at zero flow rate, and shall be capable of pumping at a rate fast enough to fill the hole in sufficient time to allow for a packer to be installed and the hole pressurized prior to the grout achieving initial set. For neat cement grout, the pump shall have a screen with 0.125-inch maximum clearance to sieve the grout before being introduced into the pump. Screens are not required for shear type mixers. The pump shall be capable of pumping both neat cement grout mixes and sanded grout mixes. The pumping equipment shall have a digital pressure gauge capable of measuring pressures of at least 100 psi or twice the required grout pressure, whichever is greater, but shall be able to measure grouting pressure to the nearest 1 psi.

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2.3.2.3 Packers

The type of single packer(s) used to isolate the stage to be ~~watertightness tested and/or~~ grouted and its means of inflation, sealing, and retrieval upon completion are the decisions and responsibilities of the Contractor. The packer shall have a minimum working pressure of 500 psi. The packer shall also be of a length sufficient to allow for proper sealing in a potentially rough sided borehole. Prior to the commencement of ~~watertightness testing and/or~~ pre-grouting, and at any time requested by the Contracting Officer, the Contractor shall demonstrate that the packer system selected is capable of sealing the hole at the maximum pressure anticipated to occur below the packer in any borehole. This shall be done by inserting the packer in a steel pipe with one closed end. The section of the pipe between the closed end and the packer shall then be pressurized with water to twice the maximum pressure differential anticipated below the packer. The system shall be monitored for leaks for a minimum of 10 minutes. The Contractor shall replace the packer if any leaks are observed at any time. The Contractor is responsible for ensuring that the test apparatus is properly designed, fabricated, and operated.
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2.3.2.4 Water Pump

If watertightness testing is conducted, the watertightness pump shall be of the progressive cavity type, and shall be capable of pumping at least 50 gallons per minute and must be capable of being regulated for flows less than 1 gallon per minute.

2.3.3 Alignment Checking Equipment

Alignment of holes drilled using non-directional equipment shall be checked by means of a rate gyrocompass, or equal equipment every 5 feet of coring advancement. For boreholes drilled with directional drilling equipment, a survey is required once the hole is completed. The survey method chosen shall be capable of accurately surveying to 1 in 700 for a vertical hole and 1 in 300 for a 5 or 15 degree hole. The survey equipment shall be capable of supplying a plot of the hole versus the requested alignment. All equipment for checking alignment of anchor holes shall be operated by personnel experienced in the operation of such equipment. The Contractor shall supply the Contracting Officer a copy of the survey plot(s) immediately upon completion of each survey and all surveys shall be submitted to the Contracting Officer, once each hole has been fully reamed, cleaned out, and prepared for anchor installation.

2.3.4 Stressing/Testing Equipment

Stressing/testing equipment shall consist of a hydraulic jack with pump for applying the load, calibrated pressure gauges and two dial gauges to measure anchor movement.

The hydraulic pumps shall be pressure compensated variable displacement and or have closed-loop control with a proportional relief valve and pressure transducer. The hydraulic pump systems shall be capable of producing the loads called out in the performance and proof tests and autonomously holding the creep test load with an accuracy of 50 psi without manual intervention. The hydraulic pump system shall be capable of running manually and automatically. The hydraulic pump system shall be programmable with user-defined target pressure, ramp rates and dwell times. The hydraulic pump system shall be capable of applying each load increment in less than 60 seconds. The hydraulic pump system shall have a digital display, showing at a minimum, test pressure (psi, ramp time (secs), and hold time (secs /minutes). The hydraulic pump system shall have a data acquisition system, including at a minimum, hydraulic pressure (psi) and time (minutes), and be downloadable thru a USB interface. The hydraulic pump system shall have an oil reservoir and cooler sized to maintain optimum oil temperature.

Stressing (jacks) shall be hydraulically operated and shall have a capacity sufficient to stress the anchors to the specified required Test Loads within the rated capacity in one stroke. The ram travel of the stressing jack shall be not less than the theoretical elastic elongation of the total anchor length at the Maximum Test Load. Stacking jacks or any multiple jack configurations for stressing an anchor is prohibited. No more than four months before a stressing jack is scheduled to be used the jack cylinder breakaway pressure shall be tested. The breakaway test shall be conducted with the jack in the horizontal position. The test shall be conducted with compressed air after all oil is removed. Compressed air shall be introduced into the caped end of the cylinder and the extending end shall be open to the atmosphere. The pressure required to begin moving the jack cylinder shall be recorded and shall be no more than 20 psi on any test. The test shall be performed at three cylinder extensions, out 2 inches, out halfway, out full extension minus 2 inches. (Example 18-inch stroke jack, tests at 2-inch, 9-inch and 16-inch cylinder stick out) The test shall be run three times at each cylinder extension any test exceeding 20 psi shall fail the jack. A failed jack may be rebuilt with new seals and retested. The jack breakaway test results shall be submitted to the Contracting Officer for review.

The equipment shall permit stressing of the tendon in increments and raising or lowering the load in the tendon. The equipment shall be calibrated with an accuracy of +/-2 percent and the calibration certificate and graphs shall be available at the site. The Contractor shall use three production gauges, to monitor hydraulic pressure during each anchor test. The gauges shall be positioned side-by-side and shall be located near the dial gauge used to measure anchor movement. During testing the high pressure side (HPS) of the hydraulic system shall have two gauges and the low pressure side (LPS) shall have one. One HPS production gauge shall be a digital gauge capable of reading to 1 psi with a minimum 0.05 percent accuracy, temperature compensated, and displays ambient temperature. The second HPS production gauge shall be a dial type with a minimum 0.5 percent accuracy and having graduations of 100 psi. The dial pressure gauge's capacity shall be such that a minimum 3/4 sweep of the gauge face shall be used at the Maximum Test Load. The digital gauge shall be used as the official testing load, the dial gauge will be used to visually help reach test loads. The third production LPS gauge shall be a dial type with a minimum 0.5 percent accuracy and having graduations of 50 psi. The Contractor shall also have a minimum of one additional dial gauge and one additional digital gauge, both calibrated with the jack and other gauges, at the project which can be plumbed into the system for a periodic verification of the production gauges. This check shall be conducted on at least one anchor test per every ten anchor tests. If the digital production gauge is off by more than 30 psi, or the dial gauge is off by more than 50 psi, at any test load, the entire system will be sent for recalibration even if the maximum 180 day calibration period has not expired.

Contractor to provide stressing equipment calibration records, pressure gage calibration records, and torque wrench calibration records prior to commencing work. Submit Pull Test Jack Certification, at least 30 calendar days prior to conducting any pull tests, load calibration certification for jacks and pressure calibration certifications for pressure gauges to be used for conducting pull testing and anchorage verification testing of embedded anchors. Re-calibrate jacks and gauges and submit re-certification for jacks and gauges when directed by the Contracting Officer. Pressure Gage Calibration results for each shift during which horizontal anchors are tested and a certification that all pressure gages used in the installation have been calibrated and found to be accurate to within 5 percent tolerances prior to use. Submit Torque Wrench Calibration, at least 14 calendar days prior to use, calibration certification for torque wrenches to be used for tensioning of anchors. Re-calibrate torque wrenches and submit re-certification for torque wrenches when directed by the Contracting Officer.

In addition to demonstrating meeting the +/-2 percent performance criteria, the stressing/testing equipment calibration shall include the following at a minimum:

(1) Independent calibration of each gauge for the stressing/testing equipment against deadweight tester, including making any adjustment if found out-of-manufacturer's stated tolerance and re-calibration. Note that this is in addition to any manufacturer's certificate of calibration at the point of sale. The gauge calibration shall be performed by a party qualified to do the calibrations and make any necessary adjustments. Gauges that cannot be made to be within tolerance shall be replaced with an equivalent and subject to calibration.

(2) Each gauge shall be calibrated using a minimum of ten load increments that represent the anticipated anchor test pressures the gauge will experience with a minimum of one increment above the maximum anticipated pressure for anchor testing and nine increments below.

(3) Minimum of a total of three runs of incremental loading, reading, and recording at three differing extensions. Ram extension for the runs shall correspond to a lower, approximate middle, and higher extension of the ram.

(4) Certification of the calibration of the gauges and stressing/testing equipment.

The Contractor shall submit complete schematics of the approved stressing/testing equipment systems showing all fluid circuits including but not limited to, miscellaneous fluid elements, pressure control valves, flow control valves, actuators and directional control valves. The operating instructions of both the hydraulic pump and stressing ram shall also be submitted. Prior to testing the 1st anchor the Contractor shall demonstrate and familiarize the Contracting Officer with the operation of the approved stressing/testing equipment systems.

Two dial indicator gauges or other device(s) approved by the Contracting Officer shall be provided to measure total tendon elongation at each load increment to the nearest 0.001-inch. The dial indicator gauges shall be capable of measuring the entire anchor movement without being reset. The

Contractor shall set up the gauges 180 degrees apart, record the readings from both and use the average of the two to evaluate anchor performance. Calibration of all dial indicator gauges shall be done using USBR 1007-89 and shall be verified no more than 30 calendar days prior to commencing work under this contract and at six-month intervals throughout the period of use. The movement measuring device shall have a minimum travel equal to 1.25 times the theoretical elastic elongation of the total anchor length at the Maximum Test Load without resetting the device.

2.3.5 Borehole Camera

The Contractor shall obtain a borehole camera system for the use on this project. The borehole camera system shall be of sufficient design and capacity to provide color images in real time and on recorded medium. The recorded medium should be in a format (.mp4) that can be viewed on all major PC operating systems. The equipment shall be compatible and able to be centralized in borehole sizes 3 inches to 15 inch diameter, and be able to be used in all inclinations required for the work in this contract. The equipment shall be equipped with two cameras in a single housing, each with a wide-angle lens for viewing downhole and side view images in wet or dry boreholes. The borehole camera must be capable of recording and viewing both cameras simultaneously. The borehole camera system shall contain automated control for either downhole or side-view video profile. The side-view lens shall include 360-degree rotation to the right and the left with no external moving parts. The camera systems shall contain low light level charge coupled device (CCD) sensors that allow the cameras to detect features such as cracks in concrete, as well as joints and discontinuities in the rock mass and the degree of joint filling with light power as low as 1 LUX. The camera system shall have the ability to increase and decrease the camera's lighting on demand. The camera system should be portable and come with a protective case for storage. Borehole recording shall include the date and time of the imaging, hole number, orientation, and depth in feet. The recording shall be completed in such a manner that the depth of features in the borehole video can be observed on the recording. Depth to structural cracks in the concrete will be recorded on the drilling log prepared during coring and compared with the depth indicated by the cores. Differentials of greater than 6 inches between the two methods shall be verified with the camera again. A final concensus depth to the deepest structural crack will be annotated on the drilling log.

2.4 GROUT REQUIREMENTS

2.4.1 Cement

The cement used in the grout shall conform to ASTM C150/C150M, Type I/II or Type 1L.

2.4.2 Water

Water shall be fresh, clean, potable, and free from injurious amounts of sewage, oil, acid, alkali, salts, or organic matter. Do not use the river water. It is known to be acidic due to upstream mining activities.

2.4.3 Admixtures for Pre-grouting

Depending on the amount of water encountered during drilling, and on the grout injection characteristics during the pre-grouting operation, other materials approved by the Contracting Officer may be required in any one hole during pre-grouting.

Admixtures which control bleed, improve flowability, reduce water content and retard set may be used in the grout subject to the approval of the Contracting Officer. Any admixtures used shall be compatible and non-detrimental with the high strength steel bar and shall be mixed in accordance with the manufacturer's recommendations. The use of bentonite as a thixotropic agent is not permitted. The use of accelerators is not permitted.

Note that aggregate and/or admixtures are anticipated for possible use only in pre-grouting operations, together with cement and water. Anchor grout for tendons is anticipated to consist only of cement and water, although admixtures intended to improve rheological properties of the grout may be used, subject to the approval of the Contracting Officer.

2.4.3.1 Superplasticizer

A high-range water reducer conforming to ASTM C494/C494M, Type A or F that will decrease the water demand at least 12 percent, may be used. A Naphthalene sulphonate based product shall be used. The material shall be shipped, stored, and mixed so as to avoid damage or waste, in accordance with the manufacturer's instructions.

2.4.3.2 Viscosifier

Natural soluble, high molecular weight biopolymer, such as Whelan Gum (or equivalent) which enhances the stability of the suspension grouts shall be used. The material shall be shipped, stored, and mixed to avoid damage or waste, in accordance with the manufacturer's instructions.

2.4.4 Aggregates (Sand) for Pre-grouting

Fine aggregate for sand-cement grout must conform to ACI 301 and ASTM C33/C33M for grout for backfilling holes. Aggregates must not contain substances which may be deleteriously reactive with alkalis in the cement.

a. The sand for grout shall be clean and consist of natural, hard, tough, durable, uncoated particles with no more than 5 percent passing the #200 sieve. The shape of the particles shall be generally rounded or cubical. The sand shall be generally well-graded from fine to coarse in accordance with ASTM C136/C136M with 100 percent passing the No. 8 sieve.

b. The sand shall be subjected to such tests as are necessary to determine its acceptability. All sampling of sand shall be in accordance with the applicable sampling provisions contained in COE CRD-C 100, provided on site, or as directed by the Contracting Officer.

c. Sand shall be stored in such a manner as to prevent the inclusion of any foreign materials in the grout. All sand shall remain in free draining storage for at least 72 hours prior to use. The percentage of surface moisture in terms of the saturated surface-dried sand will be determined in accordance with ASTM C70, or other method giving comparable results.

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2.4.5 Anchor Grout for Tendons

Cement grout mixture proportions shall be the responsibility of the Contractor. Grout for grouting tendons shall consist of a homogeneous, pumpable, ***non-shrink,*** stable mixture of portland cement (Type I/II or Type 1L) and water. The Contractor shall submit his proposed mix design to the Contracting Officer for approval. For Type I/II cement, the water content shall be the minimum necessary for proper placement but the water-cement ratio shall not exceed 0.45 by weight. For Type 1L cement, the water content shall be the minimum necessary for proper placement but the water-cement ratio shall not exceed ~~0.43~~***0.40*** by weight. Final proportions of materials shall be based on results of tests made on sample mixtures of grout. The minimum compressive strength of two-inch cubes, molded, cured, and tested in accordance with ASTM C109/C109M, shall be a minimum of 5,000 psi at the time of stressing. The Contractor shall be responsible for taking, curing, and breaking of grout test cubes for determining mix design acceptability and for quality control testing, and all sampling and testing shall be done by an independent, Engineer Research and Development Center (ERDC) Validated laboratory approved by the Contracting Officer.

2.4.5.1 Anchor Grout Mixture Proportions

At least thirty calendar days prior to installation of anchors, the Contractor shall submit the mixture proportions that will produce anchor grout of the quality required. Applicable test reports shall be submitted to verify that the grout mixture proportions selected will produce grout of the quality specified.

2.4.5.2 Polyester Resin (Epoxy) Grout

Polyester resin grout should not be used for anchors installed in wet holes. Single stage grouting can be accomplished with polyester resin grout by using fast setting resin grout in the bond zone and slower setting resin grout in the free stressing zone. The cure times of the resin grout will be affected by ground or rock ambient temperatures. Polyester resin grout must consist of high strength, unsaturated polyester resin filled with nonreactive, inorganic aggregate and a separated catalyst contained in a tube of polyester film or glass. Gel time and cure time must be appropriate for the installation procedures. The polyester resin grout must have the following minimum properties:

Compressive Strength	***12000 psi***
Tensile Strength	***4000 psi***
Shear Strength	***3000 psi***

Resin cartridges with expired shelf life are not allowed.

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2.4.6 Grout for Pre-grouting

Grouts for pre-grouting shall be based on Portland Cement Type I/II (or Type 1L) and water. Depending on the amount of water encountered during drilling, the type of grouting required, and on the grout acceptance characteristics during the pre-grouting operation, other materials may be required in any one hole. These materials will permit the Contractor to change the rheology and/or hydration characteristics of the grout in order to limit or increase grout travel. Such additional materials may include fine sand and/or viscosifier. The use of bentonite as a thixotropic agent will not be permitted. Selections and progression of the mixes for pre-grouting may be varied based on encountered conditions and project performance, as directed by the Contracting Officer's Representative.

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2.4.6.1 Grout for Tremie Grouting

Contractor to provide proposed mix for approval. The starting mix for tremie grouting will be comprised of a water:cement ratio by weight of 1.00:1.00. If ***contractor elects to conduct non-mandatory*** watertightness testing ~~is conducted~~, mix composition may vary as follows. ~~Otherwise, contractor to provide proposed mix for approval.~~

Mix	**Watertightness Test Take (gal)**	**Mix Design Water:Cement Ratio by weight**
T1	<50 gal (Note 1)	1.00
T2	<50 gal (Note 1)	0.66
T3	50 gal < x < 500 gal	0.50 + viscosifier
T4	>500 gal	0.50 + viscosifier + sand

Note 1 The Contracting Officer's Representative may direct that any mix be used when encountered conditions or project performance dictate its use is necessary.

2.4.6.2 Grout for Pressure Grouting

Contractor to provide proposed mix for approval. The starting mix for pressure grouting will comprise a water:cement ratio by weight of 0.66. If ***contractor elects to conduct non-mandatory*** watertightness testing ~~is conducted~~, mix composition may vary as follows. ~~Otherwise, contractor to provide proposed mix for approval.~~

Mix	Starting Mix Based on Watertightness Test Take (gal)	Mix Progression based on Ratio of Volume of Grout Take to Theoretical Volume of Borehole Stage	Mix Design (Water:Cement Ratio by Weight)
P1	< 50 gal	1.0-2.0 times	0.66
P2	50 gal < x < 500 gal	0.5-1.5 times	0.50 + viscosifier
P3	> 500 gal	1.5-2.0 times	0.50 + viscosifier + sand

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2.4.6.3 Pre-Grout Mixture Proportions

At least thirty calendar days prior to any pre-grouting, the Contractor shall submit the mixture proportions that will produce pre-grout of the quality required. Applicable test reports shall be submitted to verify that the grout mixture proportions selected will produce grout of the quality specified.

2.4.7 Non-shrink Grout

Grout for anchor leveling bearing plates shall be non-shrink grout conforming to ASTM C1107/C1107M. The minimum compressive strength of two-inch cubes, molded, cured, and tested in accordance with ASTM C109/C109M, shall be 5,000 psi at the time of anchor installation.

2.4.8 Grout for Waterproofing or Backfilling Holes

Grout for waterproofing boreholes which continue to have leakage after initial grouting, and for backfilling holes which are abandoned, must consist of a mixture of portland cement, masonry sand, and water. The grout mix proportions are the responsibility of the Contractor. Submit the proposed mix design to the Contracting Officer. The water content must be the minimum necessary for proper placement. Base the final proportions of materials on results of tests made on sample mixtures of grout. The minimum 28-day compressive strength of two-inch cubes, molded, cured, and tested in accordance with ASTM C109/C109M, must be 4,000 psi. The Contractor is responsible for taking, curing, and breaking of grout test cubes for determining mix design, and all testing must be done by an independent laboratory approved by the Contracting Officer. Replicate field conditions and temperatures in the curing process.

2.5 NON-SHRINK CONCRETE FOR ANCHOR RECESS

The anchors shall have their recesses filled with non-shrink concrete as specified in 03 30 53 CAST-IN-PLACE CONCRETE.

2.6 TENDON FABRICATION

2.6.1 General

Fabrication of the anchors must be as recommended by the suppliers. Completely assemble anchors with all encapsulation, centralizers, grout, and vent tubes and corrosion protection prior to insertion into the hole. Protect, transport and store fabricated anchors in a manner to prevent contamination or damage to any components.

2.6.2 Tendon

Tendon material must be unblemished and free of pitting, nicks, grease, or injurious defects. When required to maintain the tendon location within the hole, provide centralizers at a maximum of 10 foot intervals center-to-center throughout the anchor length. The entire bond length of the tendon must be free of dirt, lubricants, loose rust, corrosion-inhibiting coatings or other contaminants.

2.6.3 Bond Breaker

Bond breaker for free stressing length of unbonded anchors must consist of smooth polyethylene tubing, minimum wall thickness 0.04 inch, or smooth PVC tubing, minimum wall thickness 0.04 inch.

2.6.4 Vent Tubes

Vent tubes used during grouting operations, if necessary, must be any appropriate type for the job, as recommended by the supplier of the anchors.

2.6.5 Grout Tubes

Grout tubes must be polyethylene tubing or as recommended by the anchor manufacturer and approved by the Contracting Officer. Inside diameter of grout tubes must be adequate to fully grout the entire hole.

2.6.6 Corrosion Protection

Corrosion protection must be as indicated in the plans and specifications. Provide corrosion protection for the entire anchor and include anchorage covers and trumpets filled with corrosion inhibiting compound or grout and encapsulation of the free stressing length and bond length.

2.6.6.1 Anchorage Protection

The trumpet must be sealed to the bearing plate and must overlap the free stressing length encapsulation by at least 4 inches. The trumpet and anchorage cover must be completely filled with corrosion inhibiting compound or grout. Compound filled trumpets must have a permanent seal between the trumpet and the free length corrosion protection.

2.6.6.2 Free Stressing Length Encapsulation

Encapsulation for free stressing length must consist of a sheath of smooth polyethylene tubing, minimum wall thickness 0.06 inch; smooth polypropylene tubing, minimum wall thickness 0.06 inch; smooth PVC tubing, minimum wall thickness 0.04 inch; or corrugated tubing conforming to paragraph Bond Length Encapsulation. Sheath for bars may be heat shrinkable sleeve with a minimum thickness of 0.024 inch. Free stressing length encapsulation must extend at least 4 inches into the trumpet, but must not contact the bearing plate during testing and stressing of the tendon. Where corrugated tubing is used for sheath for unbonded anchors, a separate bond breaker must be provided.

2.6.6.3 Bond Length Encapsulation

Bond length encapsulation must consist of corrugated polyethylene tubing, minimum wall thickness 0.060 inch.

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2.7 TESTS, INSPECTIONS, AND VERIFICATIONS

~~Grout shall be sampled and tested in accordance with ASTM C109/C109M by a ERDC validated laboratory. These tests shall be at the Contractor's expense. Grout shall be tested at the minimum frequency listed in the table below.~~

~~Type of Grout~~	~~Minimum Sample Frequency~~
~~Daily Grout, Batching~~	~~One sample from each plant used per day and for each batching event.~~
~~Bond Zone (inside and outside of corr.)~~	~~For each anchor one sample taken during any placement of grout on any part of the bond zone inside and outside of the corrugated tubing for each anchor, (i.e. if grout is to be placed on any part of the bond zone on 4 adjacent anchors, a grout sample that represents each location would be required then 4 sample of grout would be required. If there are multiple placements of grout on the bond zone for a given anchor then a sample would be required for each placement.)~~
~~Corrugated Grout (outside corr. & between bond and trumpet)~~	~~One sample from each batching of grout from which grout is to be placed on the outside of the corrugated tubing (other than bond)~~
~~Second Stage Grout (inside corr.)~~	~~One sample from each batching of grout from which grout is to be placed in the second stage grout (other than redundant zones), in both the first part of the shift and the second part of the shift.~~
~~Anchor Hole Backfill~~	~~One sample from each batching of grout from which grout is to be placed for anchor hole backfill.~~
~~Leveling Pad Non-Shrink Grout~~	~~One sample from every anchor placement of non-shrink grout for leveling pads.~~

~~The Contractor shall maintain a table listing each grout sample and all types of grouting and locations represented by that sample. The Contractor shall also maintain a table detailing the timing of leveling pad non-shrink grout placements for each anchor. These tables shall be updated daily and a hard copy and electronic copy of each shall be provided to the Contracting Officer weekly. The Bond Zone Grout test results shall be provided to the Contracting Officer a minimum of twenty-four (24) hours prior to stressing. All quality control grout test results shall be submitted to the Contracting Officer at a minimum on a weekly basis.~~

2.7.1 High-Strengh Steel Bar

The Contractor shall have required material tests performed on high-strength steel bar and accessories to demonstrate that the materials are in conformance with the specifications. High-strength steel bar test results shall be furnished prior to beginning the fabrication of any anchors with steel from a given lot.

2.7.2 Encapsulation Tubing

Full anchor length encapsulation tubing rolls shall each be tagged from the factory with the manufactured minimum wall thickness, percentage of virgin resin used in its manufacturing, and a unique roll ID number. If the properties on the tag do not meet the requirements defined in ~~Paragraph FULL ANCHOR LENGTH ENCAPSULATION~~***Section CORROSION PROTECTION***, the Contractor shall reject the entire roll of encapsulation tubing.

Encapsulation tubing rolls not rejected shall be Quality Control (QC) tested for conformance to the specifications. This test shall be by taking two (2) random QC samples (one (1) sample equals a minimum 6-inch length of encapsulation tubing) from each roll of corrugated tubing. Samples shall be spaced a minimum of 50 feet apart and shall be taken in the presence of a representative of the USACE. Each sample shall be halved creating two (2) 3-inch samples (A and B). The samples shall be tagged with the sample number, sample letter (A or B) unique ID number from the roll, and location of sample in feet. The B samples shall be submitted to the Contracting Officer for Quality Assurance (QA) testing. The Contractor shall cut each sample into equal quadrants, each quadrant shall have the thickness of the tubing measured with a ball anvil and spindle micrometer in at least four (4 locations, locations shall be those that visually appear the thinnest. The diameter of the ball anvil and spindle shall be small enough as to not bridge the thin portions of the corrugated tubing giving artificially high numbers and shall have an accuracy of ±.0002 inch. The failure of any measurement to meet the minimum thickness called out in ~~Paragraph FULL ANCHOR LENGTH ENCAPSULATION~~***Section CORROSION PROTECTION*** shall result in the Contractor rejecting the entire roll of encapsulation tubing with the same unique ID. The Contracting Officer shall be notified in writing of all corrugated tubing rejected for any reason. All quality control test results shall be submitted to the Contracting Officer at a minimum on a weekly basis.

If Government QA testing finds samples not meeting the standard, the Contractor shall reject the entire roll of encapsulation tubing with that unique ID. The Government shall have a minimum of five (5) business days to perform its testing before the tubing is utilized. The Government reserves the right to obtain supplemental QA samples at any time from any encapsulation tubing.
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2.7.3 Grout Test Reports

Grout shall be sampled and tested in accordance with ASTM C109/C109M by a ERDC validated laboratory. These tests shall be at the Contractor's expense. Grout shall be tested at the minimum frequency listed in the table below.

Type of Grout	Minimum Sample Frequency
Daily Grout, Batching	One sample from each plant used per day and for each batching event.

Type of Grout	Minimum Sample Frequency
Bond Zone (inside and outside of corr.)	For each anchor one sample taken during any placement of grout on any part of the bond zone inside and outside of the corrugated tubing for each anchor, (i.e. if grout is to be placed on any part of the bond zone on 4 adjacent anchors, a grout sample that represents each location would be required then 4 sample of grout would be required. If there are multiple placements of grout on the bond zone for a given anchor then a sample would be required for each placement.)
Corrugated Grout (outside corr. & between bond and trumpet)	One sample from each batching of grout from which grout is to be placed on the outside of the corrugated tubing (other than bond)
Second Stage Grout (inside corr.)	One sample from each batching of grout from which grout is to be placed in the second stage grout (other than redundant zones), in both the first part of the shift and the second part of the shift.
Anchor Hole Backfill	One sample from each batching of grout from which grout is to be placed for anchor hole backfill.
Leveling Pad Non-Shrink Grout	One sample from every anchor placement of non-shrink grout for leveling pads.

The Contractor shall maintain a table listing each grout sample and all types of grouting and locations represented by that sample. The Contractor shall also maintain a table detailing the timing of leveling pad non-shrink grout placements for each anchor. These tables shall be updated daily and a hard copy and electronic copy of each shall be provided to the Contracting Officer weekly. The Bond Zone Grout test results shall be provided to the Contracting Officer a minimum of twenty-four (24) hours prior to stressing. All quality control grout test results shall be submitted to the Contracting Officer at a minimum on a weekly basis.

PART 3 EXECUTION

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3.1 GENERAL

The top of bond zone elevations and other physical conditions indicated on the drawings are the result of designing with the existing project design drawings, ROV inspections inside the monolith culvert, inspections of the lock control tower basement and galleries, and utilizing two core borings to identify approximate location of vertical cracking in the monolith. (See also paragraph "PROJECT SITE CONDITIONS").

Drill holes at the locations, diameters, inclinations, and tolerances shown and to the depths determined by the pilot hole drilling to provide the design bond length and capacity indicated on the drawings. The locations of the holes may be changed only as approved by the Contracting

Officer. Only 25% of the total anchors may be in progress (i.e. pilot hole drilling, enlargement drilling, pre-grouting, re-drilling, or anchor installation) at any one time. Unless otherwise specified, the Contractor must use a core drilling methodology (no percussive drilling technology) to drill a pilot hole to provide core samples for logging and then enlarge the pilot hole to the required diameter. Alignment requirements are ~~1 in 200~~***shown in drawing S-131***. ~~Do not drill holes within 10 feet of a grouted hole until the grout has set at least 18 hours.~~ ***Once grout has been placed in a core hole for pre-grouting, re-drilling shall not be conducted until the grout has set at least 12 hours. Drilling may be conducted in non-adjacent holes prior to 12 hours.*** Do not perform pressure grouting and drilling simultaneously ~~within a distance of 20 feet~~***at adjacent bore holes***. Re-drilling of ***pre-***grouted holes should be no earlier than ~~18~~***12*** hours and no later than 36 hours after grout placement. ***Once grout has been placed in a core hole for anchor bond zone grouting, drilling shall not be conducted in adjacent holes until the grout has set at least 12 hours and achieved 1,500 psi compressive strength. The intent is to avoid vibration and washout damage to the bond zone grout.*** Take care while drilling to avoid damage of any kind to the existing structures. Damages of any nature will be evaluated by the Contracting Officer and repairs or replacements must be made as required. Drill holes a maximum of 2 feet beyond the required anchor bond length, but no closer than 4 feet to the riverside wall of the monolith. Top of bond zones are a minimum of 5 feet beyond the last structural crack in the concrete. Provide a temporary plug for all holes drilled if the anchor cannot be installed prior to re-opening the lock. If a plug leaks or falls out during lock operations, clean out the borehole with high pressure water at no extra cost to the government. After successful testing and acceptance, anchor heads will be finished recessed into the wall and covered with non-shrink concrete flush with the existing wall surface.

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3.2 DRILLING HOLES

3.2.1 General

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3.2.1.1 Drilling Procedure

The physical conditions indicated on the drawings are based on information gained from original design and construction drawings and specifications (see also Paragraph SITE CONDITIONS). Holes shall go through concrete and shall be drilled at the locations, alignment, inclinations, depths and diameters shown in the Contract drawings. The locations of the holes may be changed only as approved by the Contracting Officer. For non-directional drilling, holes shall be surveyed every 5 feet of advancement. Any holes (~~noncompulsory~~ pilot, enlarging, etc.), drilled out of tolerance, shall be backfilled and redrilled in accordance with the specifications and any damage to an existing anchor hole, anchor, internal lock feature, etc., shall be repaired to the acceptance of the Contracting Officer at no cost to the Government. ~~No holes shall be drilled within 10 feet of a grouted hole until the grout has set at least 18 hours.~~***Once grout has been placed in a core hole for pre-grouting, re-drilling shall not be conducted until the grout has set at least 12 hours. Drilling may be conducted in non-adjacent holes prior to 12 hours.*** Grouting and drilling shall not be simultaneously performed within a distance of 20 feet. For redrilling, grout shall set at least ~~18~~***12*** hours, but no longer than 36 hours before redrilling. ***Once grout has been placed in a core hole for anchor bond zone grouting, drilling shall not be conducted in adjacent holes until the grout has set at least 12 hours and achieved and achieved 1,500 psi compressive strength. The intent is to avoid vibration and washout damage to the bond zone grout.*** Care shall be taken while drilling to avoid damage of any kind to the existing structures. Damages of any nature will be evaluated by the Contracting Officer. The Contractor shall repair or provide replacement at no cost to the Government for any damage as determined by the Contracting Officer.

A pilot hole will be cored first through the concrete. Concrete cores will be logged and their characteristics described. In particular, changes in concrete matrix color, aggregate size and shape, cracking, staining, etc., shall be annotated on the drilling log. Depth and elevation of water encountered, or lost, inside the bore hole will be recorded. Cores containing structural cracks (not mechanical breaks) will be stored in core boxes until project completion. Contractor will verify with Government before disposing of cores. During drilling, the location of the vertical structural cracking shown on the drawings must be identified in the borehole. The concrete containing this cracking needs to be cored for forensic purposes. A camera survey will confirm the location of cracking within the borehole. The crack needs to be pre-grouted. The intent is to waterproof this crack so it does not produce water that will interfere with bond zone grouting. A secondary benefit of this pre-grouting is filling in the crack as much as possible without causing displacement of the monolith halves. If other zones of cracking are encountered, they will be pre-grouted as well. The contractor will finish drilling the entire pilot hole and enlarge it to the required diameter before pre-grouting. ~~If necessary, the top of the designed bond zone must be shifted down to start at least 5 feet past the last structural crack encountered during drilling.~~ Anchor holes (including pilot and enlargement holes) shall be drilled a maximum of 2 feet beyond the end of the required anchor bond zone shown on the drawings~~ or 2 feet past the shifted bond zone, whichever is deeper~~. ~~Regardless of the structural crack's location within a borehole, the~~***The*** bottom of the bond zone shall not be closer than 4 feet to the outer face of the monolith opposite the starting face. Once drilled, the hole shall be cleaned to the bottom. The bottom is a maximum of two feet past the designed depth of the full anchor length encapsulation and the depth shall be sounded with a weighted tape or other approved device, in the presence of the Contracting Officer's Representative. For all boreholes which are not clean to the bottom, the Contractor shall design and utilize a safe lifting system to clean the holes to the bottom.

Once a pilot hole has been drilled to its final depth, the hole shall be surveyed by the Contractor for the Contracting Officer's approval. Once the hole has been drilled or enlarged to its final diameter and depth, the hole shall again be surveyed by the Contractor for the Contracting Officer's approval. The Government reserves the right to survey any completed pilot hole or final diameter hole for quality assurance purposes. The Contractor shall assist the Government in any way needed to survey any hole, which could include leaving or removing drilling tools in the hole, cranes for access, etc. Should the Government's survey disagree with the Contractor's survey, both the Contractor and the Government shall recalibrate their survey tools and resurvey the hole. If the surveys are still in disagreement, the Contractor shall subcontract a third party survey company (at no additional cost to the Government) using a rate gyrocompass, or equivalent, survey tool. Holes, or portions of holes, which are out of alignment shall be corrected or rejected and filled with cement grout and a new hole drilled as directed by the Contracting Officer. Slight adjustments to inclinations indicated on the drawings may be required, as directed by the Contracting Officer. The Contractor shall be responsible for all drilled holes until accepted by the Contracting Officer. Holes to replace incorrectly drilled holes shall be drilled at no additional cost to the Government.

A temporary plug or seal shall be provided for the top of all holes drilled more than two (2) days prior to installation of the anchor or if anchor will not be installed before the lock is scheduled to reopen for weekend traffic. The seal shall prevent material from getting into the borehole before anchor installation.

Redrilling done as a result of pre-grouting must be conducted between ~~18~~***12*** and 36 hours following pre-grouting. The top of the grout shall be sounded with a weighted tape or other approved device in the presence of the Contracting Officer's Representative, immediately prior to redrilling.
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3.2.1.2 Drilling Through Existing Structures

Drill holes through existing structure by core drilling equipment to reduce vibrations and prevent damage to the surrounding structure. No percussion drilling is authorized. The Contractor is advised that foreign material, including metals and other materials remaining from original construction of the existing structure, may be encountered during drilling through existing structures. Also, the monolith was designed and constructed with concrete zones of varying strengths which will likely affect drilling efficiency. A section of these zones is shown in the contract drawings, but actual strengths may differ from those shown.

3.2.1.3 Excavation and Backfill

Excavation is only required for the construction of the toe block and clearance for equipment on the lock floor. After that, installation of anchors should not require excavation.

There is to be no backfilling into boreholes with materials either cored or excavated from the boreholes. No backfill will be used as part of anchor tendon installation, only using properly designed grout installed in two stages is authorized. Unless otherwise noted in these specifications or directed by the Contracting Officer, all drill holes will remain open and ready for anchor installation. Preserve all holes in good condition until final measurement and until the records and samples have been accepted. If there is to be a delay in the anchor installation and the lock is to be brought back into service, a plug will be inserted flush with the borehole entrance to prevent inundation from the lock chamber water. Otherwise, anchors shall then be installed in accordance with the specifications and drawings.

3.2.1.4 Supplemental Borings

No payment will be made for supplemental borings that are required to be re-drilled because of mechanical failure of drilling and sampling equipment, negligence on the part of the Contractor, or other preventable cause for which the Contractor is responsible.

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3.2.1.5 Borehole Camera Logging

~~Periodically during the Contract, up to fifty times, the Contracting Officer's Representative may request access to a borehole for the purpose of viewing and logging the hole with the Contractor-provided borehole camera. The Contractor shall operate the camera for Borehole Camera Logging. One copy of the recording of each borehole shall be submitted to the Contracting Officer in digital format.~~

All holes are to be logged~~ shall be selected by the Contracting Officer~~. ~~Once a hole has been selected, the~~The Contractor shall begin flushing the hole with water to remove cuttings and material suspended in the borehole water. This will be continued until the borehole side walls can be clearly observed by the camera and logged. ~~All~~A copy of all borehole camera logging shall be ~~completed in the presence of~~submitted to the Contracting Officer.

Periodically during the Contract, up to fifty times, the Contracting Officer's Representative may request access to a borehole for the purpose of viewing and logging the hole with the Contractor-provided borehole camera. The Contractor shall operate the camera for Borehole Camera Logging. One copy of the recording of each borehole shall be submitted to the Contracting Officer in digital format per paragraph Borehole Camera.

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3.2.2 Pilot Holes

Pilot holes will be drilled using either conventional or directional drilling techniques, but shall be drilled at the locations, inclinations, and depths shown in the contract drawings. The pilot hole diameter shall be sized according to the judgment of the Contractor, but should be approximately 3 to 4 inches in diameter. The locations of the holes may be changed only as approved by the Contracting Officer, but must be within one inch of the planned location. ~~The required bond length shall start a minimum of 5 feet deeper than the last structural crack identified in the borehole, which may shift the bottom of the boring deeper than shown on the drawings. With any shifts, the~~***The*** bottom of the borehole will not extend closer than 4 feet to the outer face of the monolith. Holes may be drilled a maximum of 2 feet beyond the bottom of the required anchor bond length. Holes drilled with non-directional equipment shall be surveyed after every 5 feet of advancement to ensure they are remaining properly aligned. The driller shall keep a detailed log of production rates and bit wear and this information shall be included in the driller's log. For pilot holes drilled with either directional or non-directional equipment, the hole shall be surveyed by the Contractor upon completion. In the event that a hole gets off course and is uncorrectable within tolerances, the hole shall be grouted and the grout allowed to achieve strength equal to the surrounding concrete. Holes cannot be drilled using drilling muds. Water and air are the only acceptable drilling fluids.

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3.2.3 Concrete Cores

3.2.3.1 Arrangement of Core

Provide core boxes in compliance with PART 2, paragraph CORE BOXES. Only portions of concrete cores that contain pre-existing structural cracks, not mechanical breaks, shall be boxed. Arrange all cores neatly in the partitioned boxes in the same sequence in which they occurred before removal from the hole. Facing the open box with the hinged cover above and the open box below, arrange cores in descending sequence beginning at the left end of the trough nearest the hinges and continuing in the other troughs from left to right. Place the highest part of the core in box 1, and place the lower portions of the core in the other boxes in consecutive order. Take photographs of each boxed core length and place a ruler along its length for scale. Wipe the core with a wet rag prior to photographing it. The moisture ensures better color saturation to see the details in the concrete.

3.2.3.2 Labeling, Marking and Packing Core

Label the core boxes complying with paragraph CORE BOX LABELS on the inside and outside of the top cover. In addition, mark the depths of each core run/pull with a black waterproof pen on the spacer blocks that are placed between core pulls, if applicable. When a box is full, use spacers to prevent the core from sliding around inside the box.

3.2.3.3 Disposition of Core

While onsite, protect the filled core boxes from direct sunlight, precipitation, and freezing by some form of Contracting Officer-approved shelter that allows ventilation to the boxes. Upon completion of core drilling and sampling operations, store core boxes containing cores in an area provided by the Contracting Officer near the site of drilling. Once the project is completed, all remaining core material will be removed from the site and properly disposed of.

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3.2.4 Enlarging Pilot Holes

Following successful completion of the drilling of the pilot hole, the anchor hole shall be enlarged to the specified diameter, depth, and alignment as noted in the drawings in accordance with procedures specified in Paragraphs GENERAL, DRILLING THROUGH EXISTING STRUCTURES and ~~ANCHOR HOLES~~***Drilling Procedure***. For non-directional drilling, the hole shall be surveyed every 5 feet during drilling and at the completion. For directional drilling, the hole shall be surveyed at the completion of ~~enlargment~~***enlargement***. Final diameter holes may be drilled a maximum of 2 feet beyond the required anchor bond zone. All cuttings shall be flushed out. In the event that a hole gets off course and is uncorrectable within tolerances, the hole shall be grouted and the grout allowed to achieve strength equal to the surrounding concrete. Holes cannot be drilled using drilling muds. Water and air are the only acceptable drilling fluids.

3.2.5 Pre-grouting Anchor Holes

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3.2.5.1 General

The primary purpose of pre-grouting is to seal the cracks intersected by the anchor borehole to the extent that the anchor grout will not escape from the borehole during tendon grouting operations or be disturbed by water intrusion into the borehole. The secondary purpose is to push grout into the structural cracks previously identified in the monolith. Pre-grouting will be conducted in all boreholes after the borehole has been enlarged to its required diameter. After pre-grouting, re-drill the hole after the grout has set between ~~18~~***12*** and 36 hours. Thirty-six hours is the max time to wait to redrill because of the potential for the grout to strengthen to the point of interfering with redrilling (e.g. deflecting the drill causing borehole misalignment). The grout mixtures shall be selected by the contractor with Contracting Officer approval or be as indicated in Paragraph GROUT FOR PRE-GROUTING.

The Contractor shall select appropriate models and capacities of equipment. All mixes shall be prepared in a high speed, high shear colloidal mixer, and agitated in a paddle storage tank before being pumped into the hole by an appropriate pump. The Contractor shall demonstrate its ability to mix and pump all mixes anticipated during field trials conducted prior to construction commencing.

When water is below the top of the hole, the Contractor shall utilize tremie grouting methods first using a grout mix of the contractor's design and approved by the contracting officer, or, the grout mixtures listed in Paragraph GROUT FOR PRE-GROUTING for tremie grouting. These mixes are predicated on conducting water tightness testing first. However, this testing is not a requirement.

If artesian flows are encountered in the anchor borehole, then the Contractor shall utilize pressure grouting methods using either the grout mixes listed in Paragraph GROUT FOR PRE-GROUTING for pressure grouting, or, provide a mix design for Contracting Officer Approval. Pressures used should be equivalent to the artesian pressure plus 5 psi. Pressures above this should be approved by the Contracting Officer.

~~The Contractor should anticipate the potential for adjacent anchor holes to intersect at various depths if drill tolerances are not observed. In the event holes are intersected, the Contractor shall be capable of pre-grouting and redrilling interconnected holes, as specified in the previous paragraphs. No additional compensation, beyond unit rates for contract items, shall be made when intersecting holes are drilled.~~

All pre-grouting shall be performed in the presence of the Contracting Officer's Representative.

~~A flow chart showing the relationship between drilling, pre-grouting, and redrilling is included in Paragraph DRILLING, PRE-GROUTING, AND REDRILLING.~~

a. Tremie Grouting. At all times during the pumping of grout, the mixing and pumping plant and the associated system must be able to determine and display the following information in real time so that the information can be used to guide grouting operations, and at no greater than 1 minute intervals:

.Grout composition and specific gravity.
.Grout pumping/flow rate into the hole.
.Cumulative volume pumped into the hole, and,

.Its ratio to theoretical hole volume.

All of these parameters must be recorded and submitted to the Contracting Officer for each phase of grouting. They should also be annotated on the drilling logs.

Grout shall be placed through tremie tubes. The end of the tubes shall be located at the bottom of the hole. The end of the grout tubes shall be kept below the top of the grout surface at all times while grouting the zone. Grout shall be tremie pumped into the hole until the level of the grout is at the top of the hole. The level of the grout shall be verified by sounding. Because the angle of the borehole will be shallow, the grout surface will not be perpendicular to the borehole walls and will be more difficult to sound. If the level of the grout falls 2 feet (vertical) in less than 30 minutes, additional grout shall be tremie pumped into the hole to return the grout to the top of the hole. Additions of grout shall be continued until the fall of grout is less than 2 feet (vertical) in 30 minutes without additional pumping. The grout level in the hole shall be sounded periodically throughout the 30 minute wait period to ensure that additional pumping is not required. ***At this point, the grout will be allowed to set up for 12 hours before being re-drilling for anchor installation.*** If after 30 minutes the grout continues to drain into the crack(s), the remaining grout will be allowed to set up for at least ~~18~~***12*** hours and be redrilled. Note that the level of grout in the hole may be subject to change, as directed by the Contracting Officer's Representative, based on encountered conditions and project performance~~, as directed by the Contracting Officer's Representative~~. After the borehole has been re-drilled to remove the grout, another series of tremie grouting will be done. The mixture of the grout will be changed to reduce its viscosity and better plug the crack. The same pre-grouting procedure will be followed with the new grout mix.

The depth of the hardened grout shall be obtained with a weighted tape or approved method immediately prior to redrilling. This measurement shall be completed in the presence of a representative of the Contracting Officer.

~~If watertightness testing is conducted and produces volumes within 10 minutes of less than 50 gallons, then tremie grouting will be started with Mix #T2 as defined in Paragraph GROUT FOR PRE-GROUTING, for tremie grouting. For watertightness take values less than 50 gallons, Mix #T1 may be substituted if encountered conditions or project performance dictate that it should be, or as directed by the Contracting Officer's Representative. If watertightness testing produces volumes more than 50 gallons and less than 500 gallons, then pre-grouting shall begin with Mix #T3 as defined in Paragraph GROUT FOR TREMIE GROUTING. If watertightness testing produces volumes more than 500 gallons, then pre-grouting shall begin with Mix #T4 as defined in Paragraph GROUT FOR TREMIE GROUTING.~~

In total, Tremie grouting the hole shall be repeated ~~until it is watertight~~twice ***at most***, starting with a thinner mix ***first and*** followed by a thicker mix ***if the thinner mix does not stop leaking into the crack***. However, if the ~~holes have~~hole has been pre-grouted two times and the borehole continues to have leakage after using the thicker mix, the Contracting Officer's Representative may direct that the hole be pressure grouted using the method described below.

b. Pressure Grouting. All mixes shall be prepared in a high speed, high shear colloidal mixer, and agitated in a paddle storage tank before being pumped into the hole by an appropriate pump. The Contractor shall select appropriate models and capacities of equipment in accordance with these specifications. A packer shall be inserted at the top of the hole and sealed in the borehole. This packer shall include a pipe through which a tremie pipe can be inserted to the bottom of the hole. The hole shall be tremie filled to the top of the packer pipe. The tremie filling of the hole shall be completed at a rate which will not allow the grout rheology to change significantly prior to attachment to the pressure pumping line. The pumping line shall be attached to the packer pipe and pumping started. The pumping pressure shall be carefully controlled so that the minimum effective grout pressure as calculated at the ~~top of rock~~***vertical crack*** is 0.5 psi/foot of elevation difference between the pressure gauge and the ~~top of rock~~***vertical crack***. The maximum effective grouting pressure shall be 1.0 psi/foot of elevation difference between the pressure gauge and the ~~top of rock~~***crack***, or the gauge pressure is 5 psi in excess of any artesian pressure as measured at the top of the hole whichever is the lower maximum, or as otherwise directed by the Contracting Officer's Representative. Based on conditions encountered, Contractor may propose alternative pressures for approval. The means of inflating the packer, and assuring its proper sealing during grouting, and retrieval upon completion, shall be the decision and responsibility of the Contractor. ~~When adjacent holes have interconnected with the hole being grouted, packers shall also be inserted in the top of those holes. Once grout is observed to be discharging through those packer pipes, those packers shall be closed and grouting operation continued until refusal has been achieved.~~

During the injection of grout, the mixing and pumping plant and the associated system must be able to determine and display the following information in real time so that the information can be used to guide grouting operations, and at no greater than 1 minute intervals:

.Grout injection rate.
.Cumulative volume injected, and its ratio to theoretical hole volume.
.Grout pressure at the top of the stage.
.Grout composition and specific gravity.

All of these parameters must be recorded and submitted to the Contracting Officer for each hole grouted.

Examination of the pressure-volume-time characteristics of injection shall dictate the Contractor's mix composition, as defined in Paragraph GROUT FOR PRESSURE GROUTING. ~~If watertightness testing produces volumes more than 50 gallons and less than 500 gal, then pre-grouting shall begin with Mix #P2 as defined in Paragraph GROUT FOR PRESSURE GROUTING. If the watertightness testing produces take volumes of greater than 500 gallons, then pre-grouting shall begin with Mix #P3.~~

Refusal of the stage will be defined as attainment of the maximum effective pressure at minimal take (= 0.2 gallons per minute over a 5-minute period). The grout pressure will be maintained for a further 4 hours prior to removal of the packer or packers and cessation of the pre-grouting operation.

In the event that refusal is not achieved with the final mix—(#P3) and a volume of grout equivalent to 2.0 times the hole volume of that mix has been injected, grouting shall stop and any pressure maintained for 4 hours and the hole shall be redrilled ~~and watertightness tested as specified above~~after ~~18~~***12*** hours.

If the hole has been pressure grouted to refusal twice, and the total amount of grout pumped into the hole during the second pre-grouting operation is less than 120 percent of the theoretical amount required to fill the hole, then ~~regardless of the watertightness test results,~~ the hole shall be considered "grout tight".~~ The Contractor shall proceed with installation of the corrugated tubing.~~

For pre-grouting in new concrete with artesian pressures, the Contractor shall monitor the position of the concrete within a 30 foot radius in several locations around the hole being grouted, to be selected by the Contractor and approved by the Contracting Officer, at no cost to the Government. ~~The method shall consist of tiltmeters or survey monuments, etc, as approved by the Contracting Officer. The method shall be capable of showing movement in the concrete, as a result of grouting, in increments as small as 0.1-inch.~~ This monitoring shall be on a constant basis during the grouting process. If any movement is detected, then all grouting operations shall be ceased and the Contracting Officer notified.

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3.2.6 Re-Drilling Pre-Grouted Holes

Following successful completion of the pre-grouting, the anchor hole shall be drilled out to the specified diameter, depth, and alignment as noted in the drawings in accordance with procedures specified in Paragraphs GENERAL, DRILLING THROUGH EXISTING STRUCTURES and ~~ANCHOR HOLES~~Drilling Procedure. Re-drilling must occur between ~~18~~***12*** and 36 hours following pre-grouting to ensure the grout is not too hard to interfere with drilling. Holes cannot be drilled using drilling muds. Water and air are the only acceptable drilling fluids. One final alignment survey will be conducted to ensure the re-drilled hole still meets alignment requirements.

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3.2.7 Alignment

3.2.7.1 Tolerances

The anchor hole must be located within 1 inch of the planned location. The entry angle must be within 0.29 degrees of the specified inclination shown on the drawings. The final alignment of the drilled hole must be within ~~1 in 200 or 0.29 degrees~~ ***the tolerances*** of the theoretical alignment shown on ~~the drawings~~***drawing S-131***. Check tolerance for each anchor hole. If the hole alignment is not within these tolerances, notify Contracting Officer and perform corrective measures as directed. If tolerances cannot be maintained, then notify Contracting Officer.

3.2.7.2 Alignment Check

Each pilot hole, final diameter hole, and final redrilled hole shall be checked by the Contractor for alignment as specified herein upon completion of drilling and before commencement of any other work. This means that all pilot holes, final diameter holes, and final redrilled holes must have an alignment check done and approved by the Contracting Officer.

Direction and inclination of all anchor holes shall be checked against the specified anchor centerline by the Contractor for each 5-foot interval throughout the hole, if using non-directional methods. Checking the alignment of each anchor hole shall be done by measuring the location of the actual drilled anchor hole centerline and comparing it to the specified anchor centerline location at each 5-foot interval. For directional drilling, the location of the cutting bit must be known in real time so that adjustments can be made to stay in line. It does not require a survey every 5 feet of advancement, but a survey will be done of the entire borehole once drilling is completed.

The specified anchor centerline shall consist of a single, straight, continuous line extending from the top of the hole to the required bottom elevation of the hole at the specified azimuths and inclinations shown on the drawings. The Contracting Officer shall have access to holes for alignment surveys that may include, but not be limited to, rate gyroscopes or other down-the-hole equipment. Drill rods may be required to be removed from the hole or left in place as directed by the Contracting Officer. Access to holes for alignment surveys shall be at no additional cost to the Government.

Holes, or portions of holes, which are out of alignment shall be corrected or filled with cement grout ~~having a water-cement ratio of 0.40 or sand-cement grout~~***per paragraph Grout for Waterproofing or Backfilling Holes***, and a new hole drilled as directed by the Contracting Officer. The grout shall be allowed to cure until it has achieved a strength equivalent to the surrounding concrete or rock, to avoid diversion during drilling due to strength differences. Slight adjustments to inclinations indicated on the drawings may be required, as directed by the Contracting Officer. The Contractor shall be responsible for all drilled holes until the anchor has been installed, grouted, tested, and accepted. Holes to replace incorrectly drilled holes shall be drilled at no additional cost to the Government.

3.2.7.3 Borehole Alignment Surveys

The Contractor shall submit a report showing the results of the borehole alignment surveys. The report shall include the results of the surveys of the pilot hole, the enlarged hole at final diameter, and the hole after all re-drilling. For conventional drilling, surveys will be every 5 feet of advancement. Any deviations from intended alignment shall be identified. The Contractor shall provide a review of data from each survey. This review shall be completed by a Professional Engineer ~~registered in the State of Alabama~~. The Engineer shall also provide a recommendation regarding the acceptability of each borehole with respect to alignment tolerances.

The Contractor shall also provide an electronic copy of the alignment surveys in Excel format with the 3D coordinates for each 5-foot measurement, in tabular format. That tabular data shall include:

a. The as-planned top of borehole center location coordinates
b. The as-drilled top of borehole center location coordinates
c. The as-planned depth
d. The as-drilled depth
e. The as-planned borehole center coordinates every 5 feet
f. The as-drilled borehole center coordinates every 5 feet
g. The total amount of deviation from the as-planned borehole center coordinates every 5 feet
h. The depth(s) of any deviations outside the specified tolerances

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3.2.8 Records

The Contractor shall submit driller logs and records as specified in Paragraph DRILLER LOGS. The presence of a Government inspector or the keeping of separate drilling records by the Contracting Officer shall not relieve the Contractor of the responsibility for the work specified in this paragraph. Payment will not be made for any work for which the required records have not been furnished by the Contractor.

3.2.9 Borehole Water Level Measurement

The Contractor shall determine the depth of the water in the borehole during drilling, if present. The water level shall also be measured at the completion of drilling and before beginning the pre-grouting process. Loss of water during drilling shall also be logged. Acceptable methods include a measurement with a water level indicator through the drill rods or similar means, referenced to the top of concrete at the center line of the anchor. The top of concrete at each anchor's center line shall be surveyed by the Contractor (at no additional cost to the Government) and documented in the anchor drilling log. The method of measurement should not require pulling tools. In the event artesian conditions are encountered, the Contractor will note "flowing" at the beginning of the shift. In the event of unavoidable work in the immediate vicinity or any nearby unavoidable activity that may disturb the down hole water potentiometric surface (e.g., shift work that does not allow for an undisturbed condition overnight, etc.), the Contractor should note these potential influences with the recorded measurement. The Contractor's method should minimize local water potentiometric surface disturbances prior to recording (e.g., measurement after setting overnight, negligible work in the immediate vicinity during drilling, etc.). The water level measurements shall be noted on the boring logs. The elevation of the water level shall be determined from that measurement.

3.2.10 Interception Borehole Camera Logging

Once all drilling and/or redrilling has been completed for a given anchor borehole in which any portion of the borehole passes an adjacent pre-existing anchor(s), a borehole camera video log shall be completed for the entire length of the completed borehole. This borehole camera video log shall be used to verify that no interception of adjacent anchor(s) has occurred. This borehole camera logging shall be completed in accordance with the BOREHOLE CAMERA LOGGING paragraph of these specifications. If an interception of an adjacent anchor is detected in the video, the Contractor shall immediately notify the Contracting Officer. The video log of the interception depth interval(s) shall be submitted with the anchor records in the Closeout Submittals.

3.3 INSTALLATION OF ANCHORS

3.3.1 General

The Contractor is responsible for each drilled hole until the anchor has been installed, grouted, stressed, and accepted. Holes in concrete must be cleaned by pressurized water to remove drill cuttings. If the bottom of the inserted, ungrouted anchor rod is not within 6 inches of the bottom of the drilled hole, then the anchor must be removed and the hole must be re-cleaned. If a borehole has been plugged to allow for barge traffic and gets inundated because of plug failure, the borehole must be cleaned by pressurized water and all debris removed. Installation of anchors will be done via two-stage grouting. The bond zone shall be grouted first, the tendon stressed, and if it passes, then the annulus around the tendon shall be grouted and the anchor head covered, packed with corrosion inhibiting material, and covered with non-shrink concrete. Due to the length and weight of the anchor tendons, methods need to be employed to keep from damaging the tendon encapsulation. The Contractor shall take special care with the tendon encapsulation as not to crush, pinch, punch, buckle, cut, split, tear, melt, or in any way cause a flaw that would compromise its watertightness.

3.3.2 Placing

All the equipment used in handling and placing the anchors shall be such that it does not damage or deteriorate the high strength steel bar, corrosion protection, or the anchorages. Any equipment that comes in contact with the corrosion protection (encapsulation) shall be coated or covered with a material that is softer than the encapsulation and thick enough to prevent damage to the encapsulation. Each anchor shall be inspected by a Contracting Officer's Representative prior to and during insertion into the hole. Any damage to corrosion protection shall be repaired, in accordance with PTI Rec., prior to insertion or, if determined by the Contracting Officer to be not repairable, the anchor shall be replaced at no cost to the Government. Insertion of anchors must be in accordance with PTI DC35.1.

3.3.3 Anchorage Installation

The bearing plate and nut must be installed perpendicular to the tendon, within 3 degrees, and centered on the tendon without bending of the high strength steel bar. Corrosion protection must be maintained intact at the anchorage and any damage must be repaired prior to stressing.

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3.3.4 Placing Grout

Grouting equipment must be of type and capacity required for successful installation of the embedded anchors. All anchors must be installed using a two-stage grouting work flow to ensure adequate bond zone strength and to encase the anchor. ~~The inside of the full length encapsulation shall be grouted completely prior to anchor installation.~~ Grout shall be tremie placed through tubes. The end of the tubes shall be located at the bottom of the grout placement. The end of the grout tubes shall be kept below the top of the grout surface as it fills. Maximum grout pressure is 0.5 psi/ft. The first stage grout (bond zone grout) outside of the full length encapsulation shall be placed prior to stressing the anchor. The bond zone inside of the full anchor length encapsulation may not be filled with grout prior to the insertion of the tendon. The top of the bond zone shall be started a minimum of 5 feet past the last structural crack in the concrete***, whenever possible***. If this buffer causes the bottom of the bond zone to be deepened to within 4 feet of the back face of the monolith, then the buffer may be shortened. The bond length takes precedence over this buffer length. The anchor grout in the bond zone must reach the requirements of paragraph ANCHOR GROUT FOR TENDONS before any stressing is conducted. Wait a minimum of 24 hours after placing bond zone grout to raise the lock chamber water above the elevation of the anchor to avoid inundating the freshly placed anchor grout. ~~The second stage grout inside of the full length encapsulation shall be placed after the successful completion of stressing and lock off, and acceptance of the anchor by the Contracting Officer's Representative.~~ Grouting must be performed by a method in accordance with PTI DC35.1, paragraph 7.8. Grout tubes may be left in place for both the bonded and unbonded zones. If pressure grouting is required to perform the second stage grouting, contractor will provide recommended pressures and method to pressure grout for approval prior to grouting.

Bond zone grout must cure for at least 24 hours if an anchor installation cannot be completed prior to the lock opening for weekend commercial traffic. The stressing end shall not protrude into the lock chamber and provisions need to be made to plug the anchor borehole to prevent inundation from the chamber pool.

3.4 STRESSING

3.4.1 General Requirements

After the anchor grout in the bond zone has reached the requirements of paragraph ANCHOR GROUT FOR TENDONS, the anchors must be stressed. Prior to stressing, surfaces upon which the stressing equipment is resting must be clean and the stressing equipment must be aligned as nearly with the center of the hole as possible. An Alignment Load of 5 percent of the Design Load must be applied to the anchor prior to setting dial gauges. Stress the anchor in accordance with the anchor manufacturer's recommendation, subject to the approval of the Contracting Officer. Design and Lock-off loads are given on the drawings. Determine the lock-off procedure so that the lift-off results meet the acceptance criteria specified in paragraph Acceptance. The maximum stress must never exceed 80 percent of the guaranteed ultimate strength of anchor steel. The process of stressing the anchors must be so conducted that accurate elongation of the anchor steel can at all times be recorded and compared with the computations submitted to, and accepted by the Contracting Officer. Safety precautions must be taken to prevent workers from being behind or in front of the stressing equipment during stressing. Stressing of the anchors must be performed in a sequence submitted by the Contractor for review by the Contracting Officer. All stressing must be done in the presence of a representative of the Contracting Officer. At no time during the stressing and testing of an anchor will the stressing equipment be disconnected from the temporary stressing head or anchor. ~~72 hours after all anchors have been fully tensioned, all anchors shall be checked and re-tensioned as required. Report of any required re-stressing shall be provided to contracting officer.~~

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3.4.2 Lock-off

After completion of all required tests, the load shall be returned to the Alignment Load and the optional residual movement determined. The specified Lock-off Load shall then be applied to the anchor. A lift-off test shall be made to verify the load in the anchor tendon after the tendon is locked-off and before the stressing equipment is removed. The lift-off reading shall be within five percent of the specified Lock-off load. All lift-off tests shall be done in the presence of a representative of the Contracting Officer. After lock-off and approval of the anchor tests, the stressing length and up to the trumpet seal shall be filled with grout. Grout filling the stressing length and trumpet shall be continued until the return grout has similar specific gravity to the grout as measured at the plant. After the grouting is complete, the anchor head protective cap shall be installed, the cap filled with the required corrosion inhibiting compound, the exposed surfaces painted, and the anchorage recess shall be backfilled or covered with a protective cover, as specified. .

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3.5 FIELD QUALITY CONTROL

The demonstration test anchors as shown in the drawings must be used to verify drilling depth to crack, concrete quality, and the adequacy of the Contractor's anchor installation procedures. Demonstration test anchors must pass the performance test prior to placing other anchors within the section represented by the respective demonstration test anchor. All other anchors must be proof tested. During the stressing of each anchor, a record must be kept of gage pressure and of anchor elongation at each stage of stressing to the specified test or Lock-off Load, as applicable. The Test Load must not be exceeded. Provide a qualified engineer to evaluate the anchor test results and determine the acceptability of the anchors in accordance with the criteria indicated hereunder. ***The final Anchor Report will be stamped by a Professional Engineer.*** Final acceptance of each anchor will be made by the Contracting Officer. All tests must be run in the presence of the Contracting Officer or his representative.
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3.5.1 Anchor Load Tests

Refer to SD-07 for pull test jack certification requirements. Refer to SD-09 for pull tests records. Submit Pull Test Records conducted prior to production installations as a separate report. Submit designated Pull Test Records conducted during production on applicable daily reports that document the locations and results of pull tests conducted. Submit all data in the appropriate tables of the project database in accordance with the procedures and turnover times specified in 01 33 00 SUBMITTAL PROCEDURES.

3.5.2 Proof Test

Proof test must consist of incrementally loading the anchor and will be conducted in accordance with PTI DC35.1, Paragraph 8.3.3. During the testing of each anchor, a record must be kept of gage pressure and of anchor elongation at each stage of stressing to the Test Load required by PTI DC35.1. Measurements of the elongation of the high-strength steel bars must be made in accordance with PTI DC35.1. If the total movement between the period 1 and 10 minutes at the Test Load exceeds 0.040 inch, the Test Load must be held an additional 50 minutes and the movement readings will be taken at the interval specified in PTI DC35.1, Paragraph 8.3.3. Test records, including plots and graphical analysis of test data, must be furnished upon acceptance of each proof tested anchor in accordance with paragraph SUBMITTALS. The proof test results will be compared with similar anchors in which performance tests have been performed. If any significant variation from the proof tests occurs, the Contracting Officer may require additional performance tests.

3.5.3 Performance Test

Performance test must consist of cyclically and incrementally loading and unloading the anchor, and must be conducted in accordance with PTI DC35.1, Paragraph 8.3.2. During the testing of each anchor, a record must be kept of gage pressure and of anchor elongation at each stage of stressing to each Test Load required by PTI DC35.1. Measurements of the elongation of the high-strength steel bars must be made in accordance with PTI DC35.1. If the total movement between the period 1 and 10 minutes at the Test Load exceeds 0.040 inch, the Test Load will be held an additional 50 minutes and the movement readings will be taken at the interval specified in PTI DC35.1, Paragraph 8.3.2. Test records, including plots and graphical analysis of test data, must be furnished upon acceptance of each performance tested anchor in accordance with paragraph SUBMITTALS.

3.5.4 Driller Logs

Submit the original handwritten log and one (1) copy in typed electronic format within two days of the completion of each borehole. Keep accurate driller logs and records of all work accomplished under this contract and deliver complete, legible copies of these logs and records to the Contracting Officer upon completion of the work or at such other time or times as he may direct. All such records must be preserved in good condition and order by the Contractor until they are delivered and accepted, and the Contracting Officer will have the right to examine such records at any time prior to their delivery. Separate logs must be made for each hole. Use DRILLING LOG, ENG FORM 1836 and 1836A for logs. The following information must be included on the logs or in the records for each hole:

a. Hole number or designation and elevation of top of hole.

b. Driller's Name and Core Logger's Name

c. Inclination and diameter of the hole.

d. Make and manufacturer's model designation of drilling equipment.

e. Dates and time when drilling operations were performed.

f. Time required for drilling each run.

g. Drilling fluid used.

h. Hydraulic pressure used for crowding the core barrel during coring.

i. Depths and elevations at which core was recovered or attempts made to core including top and bottom depth of each run.

j. Description by depths of each concrete type encountered as shown on Sheet G-004 of the drawings. Concrete will be described based on matrix color, and aggregate size, shape, and density. This is to identify the different zones of concrete types used in original construction. Only cracks will be described if they are not deemed to be mechanical breaks. Mechanical breaks will be annotated, but do not need to be described in detail. At a minimum, describe infilling, staining, aperture width, texture, etc. for the cracks.

k. Percentage of core recovered and concrete quality designation per run.

l. Depth and elevation of rod drops and other unusual occurrences.

m. Depth and elevation at which water is encountered.

n. Depths and elevations at which drill water is lost and regained and approximate amounts.

o. Depth and elevation of bottom of hole, determined by measuring the drill steel length and inclination.

p. Top and bottom depths at which pre-grouting was conducted to seal off cracks.

q. Grout mix and quantities (cubic feet) used in pre-grouting to seal off structural cracks.

r. Copy of final alignment survey after hole is approved for anchor installation.

3.5.5 Anchor Records

Upon completion of installation of each anchor, the anchor records must be furnished to the Contracting Officer with grouting records during drilling, report of remedial actions taken (if any),installation records, stressing records, top of bond zone elevation, bond length, free stressing length of anchor, grout mix, grouting pressure, bags of cement injected, grout volume, a report of performance test or proof test and extended creep test results, hole alignment surveys, and a survey evaluation report. The performance test, proof test and extended creep test results must include measured lengths of drill holes and anchors, the loads and elongations recorded during testing, monitoring and stressing of the anchors, and graphs of test results as specified in paragraph SUBMITTALS. In addition as-built drawings showing the completed installation of the anchors must be furnished upon completion of installation of all anchors. In addition an as-built table shall be prepared which includes:

a. The final as-drilled elevation for the center of the top of the hole
b. Final depth drilled
c. The maximum deviation from the planned centerline
d. The depth at which the maximum deviation from the planned centerline occurred
e. The general direction of the maximum deviation referenced from the planned centerline
f. The number of times the hole was pre-grouted
h. The number of grout bags used for each pre-grout
i. The total number of bags of grout pumped into the hole
j. Grout mixes

k. Grout pressures
l. Final depth of redrilling
m. Installed Anchor Total Length
n. Installed Bond Zone Length
o. Installed Encapsulated Free Stressing Length
p. Installed Redundant Bond Zone Length
q. Alignment Load Used
r. Maximum Elongation
s. Maximum Residual Elongation
t. 1-10 min Creep Test Elongation
u. 6-60 min Creep Test Elongation (if applicable)
v. Specified Lock-off Load
w. Lift-off Load
x. Temperature during Anchor Testing
y. Anchor Test Spreadsheets
z. All hole alignment surveys

3.6 ACCEPTANCE

3.6.1 General

Acceptance of anchors must be determined by the Contracting Officer. The following criteria will be used in determination of the acceptability of each anchor:

3.6.1.1 Creep

Creep movement must not exceed 0.040 inch at maximum Test Load between the period 1 and 10 minutes of the performance or proof test. If the creep movement exceeds this limit, it must not exceed 0.080 inch at the maximum Test Load between the period of 6 to 60 minutes. If the anchor does not meet this criterion, and the cause of the behavior is not investigated and explained to the satisfaction of the Contracting Officer, the anchor shall be rejected.

3.6.1.2 Movement

Apparent free length must be calculated from the observed elastic movement in accordance with PTI DC35.1, Section 8.6.2.

3.6.1.2.1 Minimum Apparent Free Length

The calculated free length must be not less than 80 percent of the designed free tendon length plus the jack length. If the anchor does not meet this criteria, the anchor must be restressed from the Alignment Load to the Test Load and the apparent free length must be recalculated. If the anchor does not meet this criteria after 3 attempts (original plus 2 restresses), the anchor will be rejected.

3.6.1.2.2 Maximum Apparent Free Length

The calculated free length must be not more than 100 percent of the designed free tendon length plus 50 percent of the bond length plus the jack length. If the anchor does not meet this criteria, and the cause of the behavior is not investigated and explained to the satisfaction of the Contracting Officer, the anchor will be rejected.

3.6.1.3 Initial Lift-Off Reading

The initial lift-off reading must be within 5 percent of the Lock-off Load. If the anchor does not meet this criteria, the anchor must be adjusted as necessary and the lift-off reading must be repeated.

3.6.2 Replacement of Rejected Anchors

Any anchor that fails the performance or proof test or is rejected by the Contracting Officer must be replaced. A replacement anchor, including a new anchor hole if necessary, must be provided by the Contractor at no expense to the Government. The location of the replacement anchor will be as determined by the Contractor with Government approval in accordance with the redesign of the anchored structure. Provide all materials, supplies, equipment, and labor necessary to provide a new anchor assembly to the satisfaction of the Contracting Officer. No drilling will be performed for a replacement anchor until the grouting of all embedded anchors within 20 feet of the replacement anchor location has been allowed to set for at least 24 hours. Payment will not be made for rejected or failed anchors. Either remove failed anchors and thoroughly ream and clear the anchor hole, or, remove the load and cut the anchor and flush. The first method is preferable if the failure type allows for it.

-- End of Section --